

Alternative-dependent diagnosability: co-selecting system architecture and monitoring suite under a detectability requirement, demonstrated on three wind-turbine generator topologies

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Abstract

Whether the failures that matter in service will be observable is rarely a criterion when the architecture of an engineered system is chosen; condition monitoring is specified later. This paper treats diagnosability as a property of the triple (architecture, control structure, observation suite) and evaluates it with one scale-free statistic, the signal-to-drift ratio: the relative change of an observable feature under a failure mode of known extent, divided by the relative change the healthy system exhibits on its own. From it follow a minimum detectable extent and a detectable share per alternative, observation suite and failure-mode class, a decision table over the required extent and false-alarm level, and a six-step procedure that co-selects architecture and monitoring suite and tests whether a monitoring design transfers across a product family. The method is demonstrated on three wind-turbine generator topologies under identical winding short-circuits on open benchmark data. The same design-level fault produces different physical severities; where a fault becomes visible depends on the machine-drive combination, and the detectability difference between two machines on one bench comes from their healthy non-stationarity, not the signature; the cheapest suite depends on the alternative and on the requirement, voltage transducers lowering the permanent-magnet machine’s detectable extent and adding nothing for the induction machine; and a learned detector transfers across architectures only when its features are referenced to each trial’s own preceding healthy behaviour. Results carry bootstrap uncertainty, are limited to one machine per alternative, and are reproducible from the public data with the released code.

Keywords: design for diagnosability, architecture selection, sensor suite design, Design for X, condition monitoring, wind-turbine generators

1 Introduction

The architecture of an engineered system is chosen early, against criteria that are easy to quantify at that stage: performance, mass, cost, efficiency, manufacturability. Whether the failures that matter in service will be *observable*—and therefore detectable before they become expensive—is rarely among those criteria. Condition monitoring is specified later, by different people, for an architecture that is already fixed, and its difficulties are treated as signal-processing problems rather than as consequences of the architectural decision. The design community has methods that reason about failure at the concept stage—function–failure mapping and functional–failure reasoning evaluate architectures by the faults they can propagate (Tumer and Stone, 2003; Stone et al., 2005; Kurtoglu et al., 2010)—and the diagnosis community has structural methods that decide, from a model, which faults a given sensor set can detect (Krysander and Frisk, 2008; Blanke et al., 2016). Neither gives the designer an empirical *magnitude* of detectability that is calibrated against the healthy behaviour of each alternative and comparable across alternatives whose signals live in different physical units, and neither poses architecture and monitoring suite as one co-selection problem. This paper does both.

The central idea is to treat diagnosability as a property of the triple (architecture, control structure, observation suite) and to evaluate it with a single scale-free statistic computed from trials—measured or simulated—in which a failure mode of known extent is introduced into an otherwise healthy system. We call the statistic the signal-to-drift ratio (SDR): the relative change of an observable feature between a reference interval and a faulted interval, divided by the size of the relative changes the same feature exhibits between adjacent reference intervals when nothing is wrong. An SDR of one marks the point at which a fault of that extent becomes distinguishable from the ordinary non-stationarity of the healthy system at a chosen per-trial false-alarm rate. From the SDR follow the quantities a designer can write requirements on: the *minimum detectable extent* of a failure-mode class for a given alternative and observation suite, and the *share* of trials above a required extent in which it is detected. Both can be compared across alternatives and traded against the cost of the observation suite; a decision table over the requirement and the false-alarm level turns them into a selection.

The motivating domain is the generator of a large wind turbine. The choice between a permanent-magnet synchronous generator (PMSG), a squirrel-cage induction generator (SCIG), a wound-field synchronous generator (WFSG) and a doubly-fed induction generator is an established architecture-level trade (Polinder et al., 2006, 2013), and electrical-subsystem failures contribute substantially to offshore downtime (Tavner et al., 2007; Carroll et al., 2016). Stator-winding insulation faults are the archetypal

incipient failure: they begin as a short-circuit across a few turns, produce subtle signatures, and evolve into machine-destroying faults if undetected (Gandhi et al., 2011; Riera-Guasp et al., 2015). A rich literature detects them in a given machine (Thomson and Fenger, 2001; Nandi et al., 2005; Bellini et al., 2008); almost none of it asks how the choice of machine changes the problem. A recent series of open benchmark datasets makes that question answerable empirically: Tominaga et al. (2025a,b, 2024) subjected a PMSG, an SCIG and a variable-speed WFSG to the *same* inter-turn and inter-winding short-circuits, inserted at the same tapped positions of the stator winding; the PMSG and SCIG share one bench and one protocol. The datasets were published for algorithm benchmarking and refrain from comparing the machines. We use them as the test-stage instantiation of a method whose object is the comparison itself.

The contributions are:

1. A definition of *alternative-dependent diagnosability* (Section 3.3) built on the SDR statistic, its calibration against a pooled no-fault null, a binomial reliability screen for observables, and three derived quantities: minimum detectable extent, detectable share above a required extent, and signature location.
2. A six-step procedure with default rules for its framing decisions, which co-selects an architecture and its observation suite under a detectability requirement through a decision table over the required extent and the false-alarm level, and which tests whether a monitoring design transfers across the alternatives of a product family.
3. A demonstration on three generator topologies under identical winding faults, with four findings that have design content: the design-level extent of a fault and its physical severity are different axes and must both be carried; the measurement group in which a fault becomes visible depends on the machine–drive combination; the cheapest adequate observation suite depends on the alternative *and* on the requirement; and a learned detector transferred across the machines only when its features were referenced to each trial’s own immediately preceding healthy behaviour, not to a commissioning baseline.
4. A sensitivity and uncertainty analysis of the tuning choices (null quantile, window length, screen), with trial-level and tap-pair-level bootstrap intervals, and open code that reproduces every result from the public data.

Section 2 positions the work; Section 3 defines the method; Section 4 describes the alternatives, the fault protocol, the observation suites and the computation; Section 5 reports the results; Section 6 draws out the design implications and the limits of what one machine per alternative supports; Section 7 concludes.

2 Background and related work

2.1 Failure and lifecycle properties at the concept stage

Systematic design methodology treats the evaluation of alternatives against lifecycle criteria as a core activity of the conceptual and embodiment phases (Pahl et al., 2007; Tomiyama et al., 2009), and the Design-for-X programme has turned a growing list of such criteria into operational methods (Holt and Barnes, 2010). Reliability is the

criterion closest to the present work. Failure mode and effects analysis remains the dominant design-stage instrument; recent work in this journal has re-founded it on design theory (Cabanes et al., 2021), extended it with causal reasoning over failure data (Liu et al., 2026), embedded it in frameworks for demonstrating high reliability (Booth and Bayford, 2026), and coupled it to probabilistic models of early design risk (Truong et al., 2026) and to surrogates of degradation in robust design (Hassani and Khodaygan, 2026). What FMEA-type methods record about *detection* is an expert rating: the “D” column of a DFMEA is assigned, not measured, and assigned for one architecture at a time.

The design community’s own line of failure reasoning at the concept stage runs from function–failure mapping (Tumer and Stone, 2003; Stone et al., 2005) to the functional failure identification and propagation framework, which evaluates conceptual architectures by simulating how faults propagate through their functional models (Kurtoglu and Tumer, 2008; Kurtoglu et al., 2010). These methods answer *which* failures an architecture can suffer and what they affect; they do not measure whether a failure of a given size would be visible to a given set of sensors, which is the question here. In the language of axiomatic design (Suh, 2001), the monitoring function and the control function of a converter-fed machine share the design parameter “current asymmetry”: the controller acts on the same quantity the monitor would like to observe, a coupling that the present method makes visible. Alternative selection is itself a methodological topic: Eaton et al. (2026) show that technical measures omitted from a selection problem change which alternative is chosen, and Park and Okudan Kremer (2019) analyse how the topology of a product family constrains its evolution. Diagnosability is, we argue, a routinely omitted technical measure, and because monitoring designs increasingly serve a family of variants (Simpson, 2004), whether a monitoring design transfers across alternatives is a family-level question. Following Blessing and Chakrabarti (2009), the demonstration in this paper is a support evaluation: it establishes that the method can be applied and what it yields, not that it improves design outcomes in practice.

2.2 Diagnosability, testability and sensor placement

Fault detection and isolation (FDI) has a mature theory of diagnosability for systems described by models. Structural analysis determines which faults can be detected and isolated from which sensor sets without numerical parameter values (Blanke et al., 2016; Krysander and Frisk, 2008; Frisk et al., 2009), and has been applied to electric-vehicle drives with a diagnosability index that ranks candidate residual generators (Zhang and Rizzoni, 2015). In electronics, testability analysis and multi-signal flow graphs answer the analogous question for test points (Simpson and Sheppard, 1994; Deb et al., 1994). Basseville (2001) gives statistical definitions of detectability and isolability for parametric models; the prognostics community has developed sensor-selection criteria and performance metrics for health management (Cheng et al., 2010; Saxena et al., 2008) and, closest to the SDR, metrics that grade candidate features by monotonicity, trendability and prognosability (Coble and Hines, 2013). Model-based approaches (Isermann, 2005; Venkatasubramanian et al., 2003) require a model whose structure is known and trusted; data-driven diagnosis (Lei et al., 2020) requires data

Table 1 Positioning of the method among design-stage failure and diagnosability methods.

Method		Input	Output	Sensing cost		Compares alternatives	alternatives
DFMEA rating	detection	expert judgement	ordinal D score	implicit		one at a time	
Function-failure mapping, FFIP (Stone et al., 2005; Kurtoglu et al., 2010)		functional model, failure knowledge	failure modes and propagation paths per concept	no		yes, qualitatively	
Structural diagnosability, sensor placement (Krysander and Frisk, 2008; Blanke et al., 2016)	diag-	model structure	detectable/isolable fault sets per sensor set	yes, sets	minimal	per model	
PHM feature suitability (Coble and Hines, 2013)	suit-	run-to-failure data	feature scores	no		no	
This paper		trials with known fault extent (measured or simulated)	MDE, detectable share and signature location per alternative and suite; decision table	yes, per suite		yes, quantitatively	quantitatively

from the very system being designed, and its transfer across systems is an active problem (Pan and Yang, 2010).

Table 1 places the method among these. It does not require a model structure, only trials with a known fault extent; it does not require a trained detector, only observable features; and it produces a magnitude—the SDR—that is comparable across features, operating points and alternatives because it is normalised by each alternative’s own healthy non-stationarity rather than by a physical unit. Statistically it is the known-onset, fixed-sample mean-shift test of Basseville and Nikiforov (1993), expressed as a standardised effect size (Section 3.2); what is new is not the test but its use as a design measure over alternatives and suites, and the co-selection it supports. It deliberately stops at detectability: isolation and prognosis presuppose it.

2.3 Condition monitoring of electrical machines

Inter-turn short-circuits in stator windings are among the most studied incipient faults in electrical machines (Gandhi et al., 2011; Riera-Guasp et al., 2015). Motor-current signature analysis (Thomson and Fenger, 2001), negative-sequence current and impedance methods (Kliman et al., 1996; Lee et al., 2003), transient models of turn faults (Tallam et al., 2002) and the review literature (Nandi et al., 2005; Bellini et al., 2008) have established the physical signatures: an asymmetry of the winding produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous reference frame. Under closed-loop current control the picture changes: the current regulator acts against the negative-sequence current it can see, so the asymmetry migrates from the currents into the voltage commands and the controller states, with the degree of migration set by the current-loop bandwidth relative to twice the electrical frequency (Tallam et al., 2003; Cruz and Cardoso, 2004). In a wound-field synchronous machine the stator negative sequence

additionally induces a component at twice the electrical frequency in the field current (Neti and Nandi, 2009), and in permanent-magnet machines with an accessible neutral the zero-sequence voltage is a sensitive indicator (Urresty et al., 2012). These facts are well known to drive specialists; they have not been turned into a measure that a design team can apply across candidate architectures. Machine-learning detectors, including our own work on multi-fault detection in permanent-magnet drives and on transfer from finite-element simulations (Canseven et al., 2026a,b; Sadık and Canseven, 2026), are typically developed for one machine; the transfer experiment in this paper is the design-level counterpart. Finite-element turn-fault models (Vaseghi et al., 2009) and digital twins (Tao et al., 2019) are the natural trial generators for a design-stage instantiation, and how such twins should be designed is an open question in this journal (Tipuric et al., 2026; Tadeja et al., 2026).

2.4 Generator topology as a wind-turbine design decision

The generator concept of a large wind turbine—direct-drive or geared; permanent-magnet, induction, wound-field or doubly-fed; full or partial power conversion—is selected early on cost, mass, efficiency and grid-compliance grounds (Polinder et al., 2006, 2013). Reliability enters the trade through failure-rate statistics of subassemblies (Tavner et al., 2007; Carroll et al., 2016), and condition monitoring is added afterwards from a catalogue of commercial systems built around vibration, oil and electrical signals (Yang et al., 2014), not through the observability of the failures in each concept. Machine design optimisation has become highly effective at the electromagnetic and thermal objectives (Bramerndorfer et al., 2018; Canseven et al., 2025); the present method adds an objective that has so far been missing from that toolbox. Two features of the industrial problem matter for what follows: a direct-drive multi-megawatt generator runs at a fundamental frequency of the order of 10 Hz, so the ratio of current-loop bandwidth to twice the electrical frequency—and with it the migration of the fault signature into the controller—differs from that of the laboratory machines; and the doubly-fed generator, whose stator is grid-connected and not current-regulated, is the case in which that migration is absent.

3 Method: alternative-dependent diagnosability

3.1 Definitions and terms

Let $\mathcal{A}$ be a set of design *alternatives*. An alternative $a \in \mathcal{A}$ is an architecture together with the control structure it implies—for a generator, the machine topology and the converter control that the topology requires. Let $\mathcal{F}$ be a set of failure-mode classes; each class $f \in \mathcal{F}$ is parameterised by an *extent* $e \in E_f$ expressed at the design level, so that the same (f, e) can be realised in every alternative. For a winding fault the extent is the fraction of a winding branch that is short-circuited; for a bearing it might be a defect size; for a structure a crack length. The design-level extent does not fix the physical severity of the event (Section 5.1), so the framing also names a *physical severity axis* on which the alternatives can be compared. Let $\mathcal{S}$ be a set of candidate *observation suites*; a suite $s \in \mathcal{S}$ is a set of measurements with a cost $c(s)$, and $\Phi(s)$ is

the set of features computable from those measurements alone. Let $\mathcal{O}$ be the operating envelope, discretised into operating points o .

A *trial* is a record of the system in alternative a at operating point o containing a reference interval R in which the system is healthy and, immediately after it, a faulted interval F in which failure mode (f, e) is present. Trials may come from a physical test (test-stage instantiation) or from a validated simulation (design-stage instantiation, Section 3.5). We use these terms throughout: alternative (not topology or machine, except when a specific machine is meant), trial, observation suite, extent (design level) and severity (physical).

3.2 The signal-to-drift ratio

The reference interval is split into two adjacent halves R_1, R_2 , with R_2 adjacent to F . For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_{R_1}, \bar{\varphi}_{R_2}, \bar{\varphi}_F$ be the feature's mean values over the intervals, and define the relative change under fault, referenced to the half adjacent to the fault,

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_{R_2}}{|\bar{\varphi}_{R_2}|}, \quad (1)$$

and the relative change between the two healthy halves,

$$\delta_\varphi^0 = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}, \quad (2)$$

which measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. Because every trial has a healthy reference interval, δ_φ^0 is available from every trial of alternative a , faulted or not, and its absolute value pooled over the N_a trials forms an empirical null distribution of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\text{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha(|\delta_\varphi^0|)}, \quad (3)$$

where q_α is the α -quantile of the pooled null. With $\alpha = 0.95$, $\text{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable interval pairs.

What the SDR is statistically.

For a roughly Gaussian null, $q_{0.95}(|\delta^0|) \approx 1.96 \sigma$, so $\text{SDR} \geq 1$ is a two-sided 5 % test on the shift of interval means with a known change time: the fixed-sample mean-shift test, i.e. the known-onset special case of the generalised likelihood ratio of Basseville and Nikiforov (1993). The SDR is therefore a trial-level standardised effect size (Cohen's d on interval means, divided by 1.96), the detectable share defined below is an empirical power, and a share at a given α is one point of a receiver operating characteristic. Three assumptions are built in: the onset is known (the trial supplies it), the whole faulted interval is averaged, and the healthy reference lies immediately before the fault. The last assumption makes the null one of sub-second non-stationarity; a reference

established at commissioning, months earlier, has a wider null, and Section 5.7 quantifies the difference on the demonstration data. Consequently the SDR characterises the simplest detector on the feature—a threshold on the relative change of interval means—and the design quantities derived from it are statements about that detector; a detector that combines features or accumulates evidence over time can detect below $\text{SDR} = 1$, which is why the oracle variant and the transfer experiment of step 6 are part of the method.

Four properties make the SDR suitable as a design measure. It is *scale-free*: a current ratio, a voltage harmonic and a controller output are compared on one axis. It is *calibrated* against each alternative’s own non-stationarity, so a noisier alternative is not credited with detectability it does not have; the price is that the numerator and the denominator must be reported separately when alternatives are compared (Section 5.2), because a quieter null raises the SDR as much as a stronger signature does. It is *per trial*, so its distribution over trials carries the dependence on operating point and extent. And it requires *no detector*, which is what allows it to be computed before any algorithm is chosen. The relative form (1) presumes a feature with a non-zero healthy level; for features whose healthy value is a noise floor an absolute-change variant with the same structure is used, and Section 5.7 shows that the two rank the features alike.

Safeguards.

Three safeguards are part of the definition. (i) *Reliability screen*. On healthy trials—trials in which the “faulted” interval contains no fault, only the switching operation that would insert one—a feature should exceed $\text{SDR} = 1$ at the nominal rate $1 - \alpha$. A feature whose count of such false alarms in the n_h healthy trials is incompatible with that rate (one-sided binomial test at the 5% level) is flagged as unreliable for that alternative and excluded from the design quantities; with $n_h = 9$ and $\alpha = 0.95$ this means three or more false alarms. Features whose null quantile exceeds one, i.e. whose healthy relative change exceeds 100%, are excluded as uninformative. (ii) *Fixed versus oracle feature*. A suite is evaluated by one *fixed* feature—the feature with the highest median SDR over the trials of a failure-mode class among the screened features of the suite, chosen once—or by the *oracle* feature, the best feature of each trial. The fixed variant is what a threshold detector achieves; the oracle variant is an upper bound for a detector that combines the suite’s features, and because it takes a maximum over features its false-alarm rate on healthy trials is reported beside it. Since the fixed feature is chosen on the same trials it is evaluated on, a *nested* variant chooses it on all but one tap pair and evaluates it on the held-out pair; both are reported. (iii) *Uncertainty*. Every design quantity carries a bootstrap interval: over tap pairs (the physical fault locations) for statements about a failure-mode class, over operating points within an extent level for the minimum detectable extent, with Wilson intervals for shares.

3.3 Derived design quantities and the definition

From the SDR distribution over trials we define, for each alternative a , suite s and failure-mode class f :

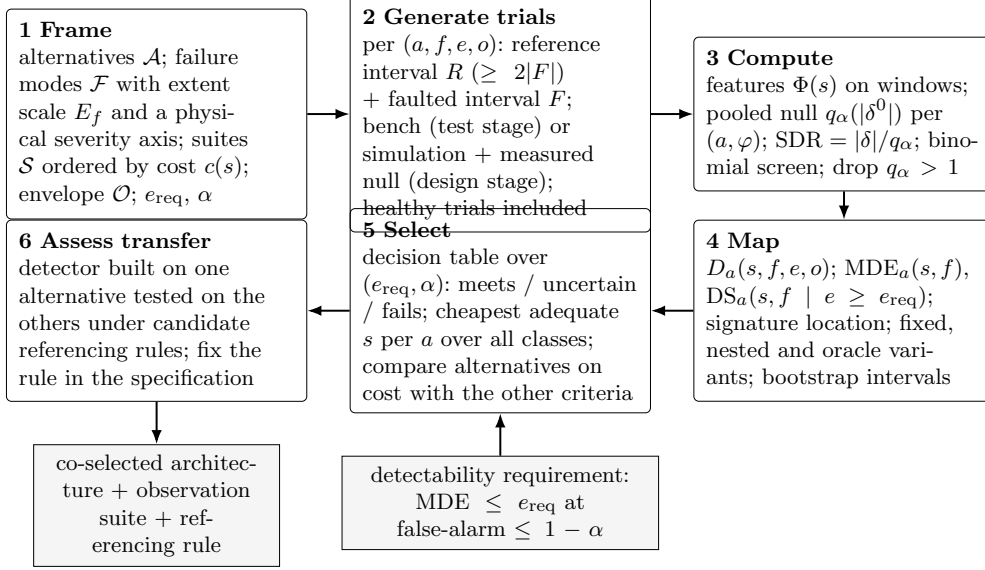


Fig. 1 The design procedure. Steps 3–4 are fully specified by Eqs. (1)–(3); steps 1, 2, 5 and 6 are design decisions for which Section 3.4 gives default rules.

- the *diagnosability map* $D_a(s, f, e, o)$: the SDR of the suite’s fixed feature at extent e and operating point o ;
- the *minimum detectable extent* $\text{MDE}_a(s, f)$: the smallest tested extent level e^* such that $\text{median}_o D_a(s, f, e, o) \geq 1$ for every tested $e \geq e^*$ (the limit-of-detection convention; equal extents realised at different winding locations form one level);
- the *detectable share* $\text{DS}_a(s, f \mid e \geq e_{\text{req}})$: the fraction of faulted trials with $e \geq e_{\text{req}}$ in which $\text{SDR} \geq 1$, a quantity written for a required extent so that it does not depend on how the test matrix samples the extent axis;
- the *signature location*: the measurement group—stator currents, terminal voltages, controller internals, mechanical quantities, excitation—whose best screened feature attains the highest SDR in the largest share of trials.

Alternative-dependent diagnosability is the map $(a, s, f) \mapsto (\text{MDE}_a, \text{DS}_a, \text{location})$. For a requirement $(f, e_{\text{req}}, \alpha)$ the *adequate* suites of alternative a are those with $\text{MDE}_a(s, f) \leq e_{\text{req}}$, and diagnosability is alternative-dependent for that requirement when the cheapest adequate suite differs between alternatives by more than the bootstrap uncertainty of the MDE. MDE and DS are the quantities against which requirements are written (“inter-turn faults of extent $\geq e_{\text{req}}$ shall be detectable over the operating envelope at a per-trial false-alarm rate $\leq 1 - \alpha$ ”); the signature location tells the designer *why* an alternative behaves as it does.

3.4 Design procedure

The procedure (Fig. 1) has six steps. Steps 3–4 are mechanical and fully specified by Eqs. (1)–(3); steps 1, 2, 5 and 6 are design decisions, for which the method supplies the following default rules.

1. **Frame.** Enumerate the alternatives $\mathcal{A}$, the failure-mode classes $\mathcal{F}$ with a design-level extent scale E_f realisable in every alternative and a physical severity axis, the candidate observation suites $\mathcal{S}$, and the operating envelope $\mathcal{O}$. Set e_{req} as the smallest extent whose growth to an unacceptable consequence is faster than the inspection or maintenance interval, and $1 - \alpha$ as the per-trial false-alarm rate the operator tolerates over that interval. Order the suites by cost; an ordinal rank by added hardware and integration is the default, a monetary cost including installation and maintenance is used when available.
2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference interval of at least twice the faulted duration, immediately followed by the faulted interval. Test at least the extent levels that bracket e_{req} and the corners of the operating envelope, and include enough healthy trials with the same timing for the screen (nine allow the binomial rule above). At the test stage the trials are prototype or benchmark experiments; at the design stage the faulted numerator is simulated and the null is supplied as described in Section 3.5.
3. **Compute.** Extract the features $\Phi(s)$ for all suites on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, apply the screen.
4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into MDE_a , DS_a and the signature location, for the fixed, nested and oracle variants, with bootstrap intervals.
5. **Select.** Build the decision table: for each alternative, class and suite, and for the candidate values of e_{req} and α , classify the suite as *meets* when the bootstrap probability that $\text{MDE} \leq e_{\text{req}}$ is at least 0.95, *fails* when it is at most 0.05, and *uncertain* otherwise. The adequate suite of an alternative is the cheapest that meets the requirement for *every* failure-mode class in $\mathcal{F}$; an uncertain verdict is resolved by more trials at the extents around e_{req} or, failing that, by specifying the next suite as a fallback. The alternatives are then compared on the cost of their adequate suites together with the other selection criteria. The architecture decision is affected when, for some alternative, no affordable suite meets the requirement; otherwise the method changes the monitoring specification, not the architecture, and says so.
6. **Assess transfer.** If the observation suite and its detector must serve a family of alternatives, test whether a detector built on one alternative’s trials detects the others’ faults under the candidate feature-referencing rules (physical units; scaled on the target’s healthy trials; referenced to each trial’s own preceding healthy interval). The rule that transfers becomes part of the monitoring specification.

3.5 Design-stage instantiation

The demonstration in this paper is a test-stage instantiation on built machines. A design-stage instantiation must supply the two ingredients of the SDR separately.

The numerator—the change of a feature under a fault of extent e —is what a finite-element or circuit model of each alternative with the fault injected can provide (Vaseghi et al., 2009). The denominator—the healthy non-stationarity of sensor noise, controller activity, thermal drift and operating-point wander—is a property of hardware and environment that such a model does not reproduce, and a simulated SDR that supplied its own null would be optimistic by an unknown factor. The null must therefore come from measured healthy records of hardware of the same class (a predecessor product, a prototype of one alternative, or a benchmark such as the one used here), transferred to the other alternatives under the assumption that the healthy non-stationarity of a feature depends on the sensing and control chain more than on the machine; that assumption is itself testable with the method, by comparing the null tables of the alternatives for which hardware exists (Table 5). The validation of a simulated instantiation is then a comparison of simulated and measured SDR on a few physical trials, which is the role the three benches would play in a design study of the next generator.

3.6 Scope

The method applies to failure-mode classes that can be seeded at a known extent, at a stationary operating point, with a healthy interval immediately before the fault, in every alternative; the extent scale must be realisable in all of them, and the physical severity axis is declared beside it. It evaluates detectability by a threshold on interval means, not the time to detection, and it stops at detection: isolation, severity estimation and remaining-life prediction presuppose detectability and can be built on the same trials. It does not model the cost of missed detection, which enters through e_{req} and α in the requirement. Run-to-failure data, in which the fault grows gradually and no healthy interval precedes a known onset, do not fit the trial structure and would need a different null.

4 Demonstration: three generator topologies under identical winding faults

4.1 Alternatives and data

The three alternatives are laboratory-scale generators of the same power class on instrumented test benches at the University of São Paulo, whose records were published as open benchmark datasets (Table 2). The PMSG (Tominaga et al., 2025a) and the SCIG (Tominaga et al., 2025b) are four-pole, 230 V, double-star machines from the same manufacturer, rated 2.5 kVA and 2.5 kW respectively, installed on the *same* bench: the same induction-motor prime mover and frequency converter, the same rapid-control-prototyping platform and field-oriented control (rotor angle from an encoder for the PMSG, from a flux observer for the SCIG), the same sensors, the same acquisition (20 kHz, 3 s records), the same fault-insertion hardware and the same test script. The variable-speed WFSG (Tominaga et al., 2024) is a 2 kVA four-salient-pole machine on an earlier bench of the same group (Rocha et al., 2023) with a three-level converter, field-oriented control with encoder angle, 4 kHz acquisition and 1.3 s records. The PMSG-SCIG pair is therefore a controlled comparison in which

Table 2 The three alternatives. PMSG and SCIG share one bench and one protocol; the WFSG bench differs where indicated. Ratings as given by the manufacturer (kVA for the synchronous machines, kW for the induction machine).

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, static exciter
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.3 s
Reference / faulted interval	0.68 s / 0.36 s	0.68 s / 0.36 s	0.49 s / 0.06–0.28 s
Fault impedance	2.6 Ω	2.6 Ω	1, 2.83, 5.66, 11.32 Ω
Operating points	3 speeds $\times$ 3 torques	same	716–1910 rpm, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials	216	216	438 (198 at 2.83 Ω)

everything but the machine and the control structure it requires is held fixed; the WFSG is a third alternative with the same fault protocol but a different bench, and is analysed with that difference kept explicit. Terminal voltages are measured at the machine terminals by each bench’s voltage transducers, whose bandwidth the data papers do not document, so the terminal-voltage features carry the sensing chain of their bench as well as the machine.

4.2 Failure-mode class and extent scale

All three machines were built with 24 tapped derivation points along their stator windings at the same relative positions, so that a short-circuit between two taps realises a fault of known extent (Table 3); the tap positions, given as percentages of a winding branch from the neutral, are identical in the three data papers, which is what makes the extent scale realisable in every alternative. Two classes are used: inter-turn short-circuits (both taps on the same branch) and inter-winding short-circuits (taps on the two parallel branches of a phase). The extent e is the fraction of a branch between the two taps; it ranges from 2.7 % to 12.1 % for the twelve inter-turn cases and from 2.3 % to 40.2 % for the twelve inter-winding cases. Extents that coincide at different winding locations (2.8 % twice, 7.4 % three times, 11.6 % twice) form one extent level. Faults were closed through a contactor and a series resistance after the machine had reached steady state: 2.6 Ω for 0.4 s on the PMSG and SCIG; on the WFSG the experimenters raised the resistance with extent (1 Ω for the inter-turn cases below 11 %, 2.83 Ω for the remaining inter-turn cases and the small inter-winding cases, 5.66 and 11.32 Ω for the large inter-winding cases), so that on that bench extent and impedance are confounded. The fault current is measured on all benches; it is used only as ground truth for the physical severity axis, never as a feature. Because the series resistance is of the order of the winding impedance of these small machines, the fault current is essentially the voltage induced in the shorted turns divided by the resistance, a *resistor-limited* severity that differs from a bolted fault, which is limited by the leakage reactance of the shorted turns (Section 5.1).

Table 3 Tap pairs and fault extents (fraction of the winding branch between the taps), identical for the three alternatives.

Inter-turn			Inter-winding		
Taps	Phase	e (%)	Taps	Phase	e (%)
D11–D12	A	2.7	D04–D06	A	2.3
D09–D10	A	2.8	D16–D18	A	3.2
D23–D24	A	2.8	D04–D07	A	9.7
D21–D22	A	3.8	D16–D19	A	10.6
D06–D07	A	7.4	D01–D06	A	14.3
D14–D15	B	7.4	D13–D18	A	14.8
D18–D19	A	7.4	D01–D07	A	21.7
D02–D03	B	7.7	D13–D19	A	22.2
D17–D20	C	11.6	D12–D21	A	33.7
D13–D16	A	11.6	D11–D21	A	36.4
D01–D04	A	12.0	D12–D22	A	37.5
D05–D08	C	12.1	D11–D22	A	40.2

The PMSG and SCIG matrix is complete: 24 fault cases $\times$ 3 speeds (1200, 1500, 1800 rpm) $\times$ 3 torques (5.2, 6.4, 8.0 N m), the torque–speed pairs scaled from the operating region of the IEA 15 MW reference turbine (Gaertner et al., 2020) to the laboratory speed range (the electrical frequency is not scaled), plus the healthy condition at each operating point; the SCIG generates with slip, at 59.6 Hz at 1800 rpm. The WFSG matrix covers the same 24 tap pairs at six speeds (716–1910 rpm) and four per-unit torque levels, sparsely and with two to six repetitions, so its operating points are not matched to the other two. The WFSG benchmark also contains phase-to-phase faults, a class absent from the PMSG/SCIG protocol; they are excluded from every analysis. The analyses use the WFSG’s 438 inter-turn and inter-winding trials at all impedances for the null, the feature ranking and the detectable shares, with the impedance as a covariate, and the 198 trials at 2.83Ω —the impedance closest to the PMSG/SCIG value, covering four inter-turn tap pairs of 11.6–12.1 % and four inter-winding pairs of 2.3–10.6 %—for the minimum detectable extent and the decision table. Each table and figure states which set it uses.

4.3 Observation suites and features

Five observation suites are considered, ordered by the data access, hardware and integration they add to a bare generator–converter set (Table 4). *Drive-internal* uses only quantities the converter controller already has: the dq currents, its own dq voltage commands, the proportional–integral controller outputs, the modulation duty cycles, the DC-link voltage, the estimated electrical torque and the speed. *3CT* uses the three phase currents—from dedicated transducers or from the converter’s own current sensors routed to a monitoring system—with angle-free features only; for a converter-fed machine its cost is data access and bandwidth rather than new hardware. *3CT + 3VT* adds three terminal voltage transducers; *3CT + 3VT + drive* combines the external transducers with controller access, which also allows the measured terminal voltages to be transformed into the synchronous frame; and *+ torque transducer* adds

Table 4 Observation suites and the features they can compute. f_e : electrical frequency; “ $2f_e$ ”: amplitude of the component at twice the electrical frequency; I_1, I_2 (V_1, V_2): positive- and negative-sequence magnitudes at f_e .

Suite	Added access / hardware	Features
Drive-internal	controller data	I_d, I_q : std, $1f_e, 2f_e$; $V_d^{\text{cmd}}, V_q^{\text{cmd}}$: $2f_e$; PI outputs d, q : std, $2f_e$; duty-cycle negative sequence D_2/D_1 ; T_e : std, $1f_e, 2f_e$; DC-link std; speed std
3 CT	three phase currents, no angle	I_2/I_1 , RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q : $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m : std, $1f_e, 2f_e$
Excitation (WFSG)	field-current measurement	I_f : std, $2f_e$

a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite), available because its exciter is static; a brushless exciter would not expose it (Neti and Nandi, 2009). Each feature is assigned to the cheapest suite whose measurements suffice to compute it. The suites are those the datasets can populate: zero-sequence voltage (which needs neutral access), stray flux, winding temperature, vibration and partial discharge are not recorded, and the angle-free Park’s-vector modulation of the currents was not computed in this version; the suite conclusions therefore hold among the suites and features considered. The zero-sequence current is not used: the machines have isolated star points, so it is a noise floor.

Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle. The electrical frequency and the phase rotation are estimated per window from the phase slope of the current space vector (falling back to the nominal value outside 0.5–1.5 times it), so that no angle sensor is needed for the external suites; sequence components are obtained by Hann-windowed projection of each phase quantity onto the fundamental and symmetrical-component transformation; $1f_e$ and $2f_e$ amplitudes of synchronous-frame quantities by projection after mean removal. On the WFSG bench the measured converter-side phase voltages serve as terminal voltages, the synchronous-frame voltages are obtained from them with the encoder angle, the commanded voltages are the controller’s, and the electrical torque is computed from power and speed. Appendix A lists the definitions.

4.4 Computation

For the PMSG and SCIG, the reference interval is 0.30–0.98 s after the start of each record, split at 0.62 s into R_1 (0.32 s) and R_2 (0.36 s); the contactor is commanded at 1.000 s and the faulted interval runs from 30 ms after the command to 10 ms before its release. The relay picks up about 85–90 ms after the command, so the faulted interval contains about 55 ms of healthy time and the insertion transient (at most 15 % of its length), which dilutes the fault-interval mean and makes the PMSG/SCIG SDRs slightly conservative. For the WFSG the fault interval is located from the measured

fault current: the first contiguous interval in which its 10 ms RMS exceeds the larger of 0.3 A and six standard deviations of the pre-fault RMS, ending at the first fall below that threshold after 50 ms; the reference interval runs from 0.05 s after the start of the record to 10 ms before the onset and is split at 0.30 s (R_1 0.25 s, R_2 about 0.24 s). Recovery windows start 0.30 s (PMSG, SCIG) or 0.15 s (WFSG) after the fault is cleared. The main PMSG–SCIG comparison uses $n_c = 5$; the three-alternative comparison uses $n_c = 3$ for all three machines because the WFSG fault intervals are short: at 23.9–63.7 Hz a faulted interval holds between zero and eleven three-cycle windows, four trials hold none and are dropped, and 81 of 438 hold fewer than three. The sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and to $\alpha \in \{0.90, 0.95, 0.99\}$ is reported, with the screen applied at each setting. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 WFSG records, every record contributing a healthy reference interval.

Uncertainty is quantified by bootstrap (Efron and Tibshirani, 1993) with 2000 resamples and a fixed seed: the median SDR and the detectable share of a failure-mode class are resampled over the twelve tap pairs, the same resampled pairs for both machines so that the SCIG/PMSG ratio is matched; the minimum detectable extent is resampled over the operating-point trials within each extent level; shares also carry Wilson intervals; the null quantile is held at its full-sample value. The decision table uses 1000 resamples per suite, alternative, class and α , with $e_{\text{req}} \in \{3, 5, 8, 12\} \%$.

Step 6 of the procedure is instantiated as a window-level change detector: a gradient-boosted tree classifier (Friedman, 2001) (histogram implementation, depth 3, 200 iterations, learning rate 0.08, fixed seed) and, for the PMSG–SCIG pair, a logistic regression, trained to separate faulted windows from healthy ones. Within an alternative the detector is evaluated by five-fold cross-validation grouped by tap pair, with the healthy trials grouped by speed so that every fold contains some, so that the tested fault locations are never seen in training; across alternatives it is trained on all trials of one machine and tested on all trials of another. Three feature-referencing rules are compared: physical units; features z -scored on the target machine’s healthy trials (a commissioning baseline); and the absolute z -score of each window against the mean and standard deviation of the *first half* of its own trial’s reference interval. Negatives are the reference and recovery windows of all trials plus the fault-interval windows of the healthy trials, the only negatives that contain the contactor operation without a fault. Because the windows used to normalise a trial are trivially separable under the third rule, every metric is reported on all windows and on the out-of-reference windows only, and the false-alarm rate of the threshold that gives 1 % false positives on all negatives is reported on the healthy trials’ fault-time windows. The true-positive rate at 1 % false-positive rate is a point of the receiver operating characteristic, not an operating threshold.

All computations are implemented in Python and released with the paper; the pipeline runs from the feature caches to every table and figure in under thirty minutes on a four-core machine. A large language model was used as an assistant in writing the analysis code and in drafting and editing the text; every analysis and every reference was verified by the author (see the declarations).

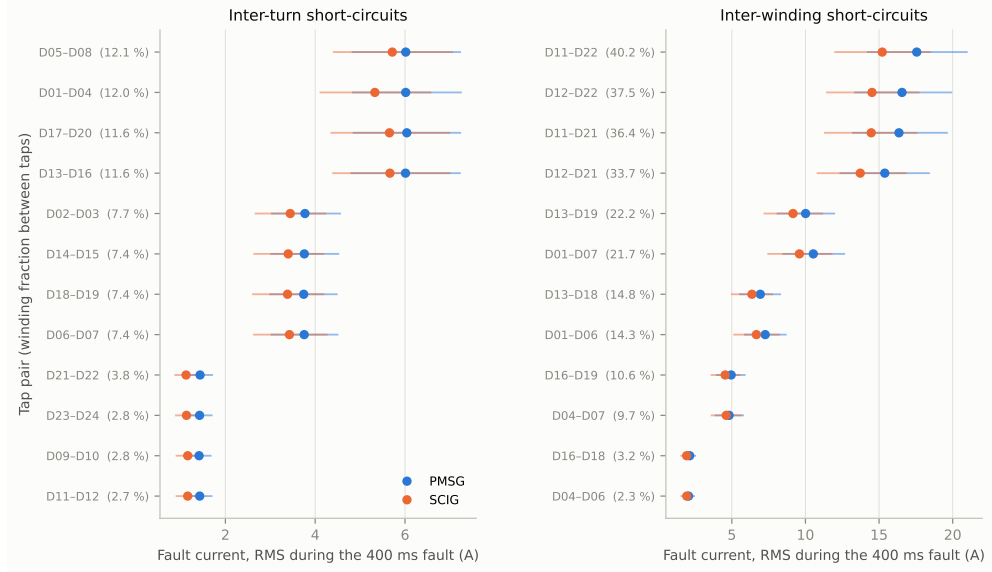


Fig. 2 Fault current per tap pair, PMSG and SCIG: mean over the nine operating points (dot) and range (bar). Rated currents 6.3 A (PMSG) and 7.7 A (SCIG).

5 Results

5.1 The same design-level fault is not the same physical fault

Figure 2 shows the fault current for each tap pair in the PMSG and the SCIG. For the same pair the SCIG’s fault current is about 10 % lower (medians 3.3 against 3.7 A for inter-turn and 7.7 against 8.5 A for inter-winding pairs), but the difference that matters for detection is relative to what the stator sensors see: the fault current is 1.45 times the pre-fault stator current (RMS to RMS, median over pairs) in the PMSG and 0.63 times in the SCIG for inter-turn faults, and 3.3 against 1.5 times for inter-winding faults, because the SCIG’s stator current carries the magnetising component (pre-fault stator current 5.3 A against 2.6 A RMS). In per-unit of rated current the inter-turn pairs give 0.60 (PMSG) and 0.44 (SCIG). The WFSG trials at the matching impedance give 1.20 (inter-turn, 11.6–12.1 % extents) and 1.03 (inter-winding, 2.3–10.6 %), and 0.65 at the lower impedance used for its small inter-turn extents. With a series resistance of the order of the winding impedance, these currents are resistor-limited: the fault current is essentially the voltage induced in the shorted turns divided by the resistance, so it scales with volts per turn and speed and differs between the machines mainly through their terminal voltage per turn. A bolted fault would be limited by the leakage reactance of the shorted turns and reach several times rated current, and—the first-order difference between the topologies—could be de-excited in the SCIG and the WFSG but not in the PMSG. A fault specified as “ e % of a winding” therefore arrives at the sensors of the three alternatives as three different physical events, and the benchmark’s emulation fixes which events; Section 5.3 uses both the design-level and the physical axis.

Table 5 Null of the signal-to-drift ratio: 95th percentile of the absolute relative change of a feature between two adjacent healthy intervals, pooled over all trials of a machine ($N = 225$ for the PMSG and the SCIG; $N = 438$ for the WFSG, all its inter-turn and inter-winding trials at every fault impedance), for five-cycle windows (PMSG, SCIG) and three-cycle windows (all three).

Feature	PMSG (5 c)	SCIG (5 c)	PMSG (3 c)	SCIG (3 c)	WFSG (3 c)
I_2/I_1	0.162	0.034	0.160	0.036	0.082
V_2/V_1	0.086	0.033	0.095	0.037	0.037
$I_d 2f_e$	0.563	0.016	0.444	0.015	0.168
$I_q 2f_e$	0.184	0.060	0.200	0.061	0.087
$V_q 2f_e$	0.165	0.060	0.176	0.061	0.099
$V_d^{\text{cmd}} 2f_e$	0.527	0.019	0.429	0.020	0.168
D_2/D_1	0.090	0.010	0.093	0.017	0.067
$I_a \text{ h3}$	0.216	0.035	0.162	0.039	0.574
$T_e 2f_e$	0.649	0.625	0.551	0.453	0.588
speed std	0.431	0.521	0.459	0.457	1.295
$I_f 2f_e$	—	—	—	—	0.339

5.2 Where the signature appears, and why the SDR differs

Table 5 gives the null of the SDR—the 95th percentile of the healthy relative change—for the main features. It differs by up to a factor of thirty between the PMSG and the SCIG, and the difference is not confined to sensed quantities: the null of the duty-cycle negative sequence, a controller output that involves no sensor, is five times larger on the PMSG. The healthy non-stationarity is a property of the machine–drive combination, not of the transducers alone. Table 6 decomposes the SCIG/PMSG ratio of median SDR into its numerator and denominator: for inter-turn faults the PMSG’s relative changes are the *larger* ones on every feature (the SCIG/PMSG ratio of median $|\delta|$ is 0.37–0.71), and the SCIG’s higher SDR is entirely a quieter null (ratio of null quantiles 2.6–35). For inter-winding faults the numerators are comparable (0.3–1.4) and the denominators again decide. The SDR is meant to combine both—a fault is detectable relative to what the healthy system does—but the comparison of alternatives must be read with the decomposition beside it.

With the null in place, Fig. 3 shows the median SDR of every feature for the three alternatives, and Table 7 the share of trials in which each measurement group carries the strongest signature. In the PMSG the terminal-voltage group is strongest in 67 % of inter-turn and 88 % of inter-winding trials, through V_2/V_1 (median SDR 3.0 and 62 at five-cycle windows; 76 and 94 % of trials detectable), followed by the duty-cycle negative sequence; the stator-current negative sequence I_2/I_1 is weak for inter-turn faults (median SDR 1.5, 56 % detectable). In the SCIG the controller group is strongest in 56 and 67 % of trials, through the $2f_e$ ripple of the d -axis current and its PI output (6.7 and 74; 79 and 100 % detectable), the stator-current group in 28–33 %, and the terminal-voltage group in 13 % and none. The two machines rank the 30 common features with a Spearman correlation of 0.70 (inter-turn) and 0.89 (inter-winding): the same features are informative in both, in a different order. In the WFSG the strongest inter-turn features are V_2/V_1 , the q -axis current ripple and the commanded q -axis voltage, and for inter-winding faults the q -axis ripple, the commanded voltage, the duty cycles and the third current harmonic; its field-current $2f_e$ ripple, an observable the other two machines do not have, detected 86 % of the inter-turn trials at the

Table 6 Decomposition of the SCIG/PMSG ratio of median SDR into its numerator (ratio of median $|\delta|$, the relative change under fault) and its denominator (ratio $q_{0.95}^{\text{PMSG}}/q_{0.95}^{\text{SCIG}}$ of the null quantiles); five-cycle windows.

Class	Feature	$ \delta $ P	$ \delta $ S	q_{95} P	q_{95} S	Num.	Den.	SDR ratio
inter-turn	V_2/V_1	0.257	0.136	0.086	0.033	0.53	2.64	1.40
inter-turn	I_2/I_1	0.238	0.152	0.162	0.034	0.64	4.82	3.08
inter-turn	$I_d 2f_e$	0.291	0.107	0.563	0.016	0.37	35.09	12.95
inter-turn	$\text{PI}_d 2f_e$	0.289	0.107	0.557	0.016	0.37	34.71	12.91
inter-turn	$V_q 2f_e$	0.222	0.155	0.165	0.060	0.70	2.73	1.91
inter-turn	$V_q^{\text{cmd}} 2f_e$	0.197	0.139	0.162	0.039	0.71	4.17	2.94
inter-turn	D_2/D_1	0.251	0.111	0.090	0.010	0.44	8.96	3.98
inter-winding	V_2/V_1	5.318	1.569	0.086	0.033	0.30	2.64	0.78
inter-winding	I_2/I_1	1.833	1.869	0.162	0.034	1.02	4.82	4.91
inter-winding	$I_d 2f_e$	2.335	1.188	0.563	0.016	0.51	35.09	17.85
inter-winding	$\text{PI}_d 2f_e$	2.344	1.188	0.557	0.016	0.51	34.71	17.60
inter-winding	$V_q 2f_e$	3.688	1.495	0.165	0.060	0.41	2.73	1.11
inter-winding	$V_q^{\text{cmd}} 2f_e$	0.845	1.184	0.162	0.039	1.40	4.17	5.84
inter-winding	D_2/D_1	2.942	0.854	0.090	0.010	0.29	8.96	2.60

Table 7 Signature location: share of fault trials (%) in which each measurement group’s best screened feature attains the highest SDR (representative feature in parentheses). PMSG and SCIG, five-cycle windows.

Machine	Class	Stator currents	Terminal voltages	Controller/dq	Mechanical
PMSG	inter-turn	8 (I_2/I_1)	67 (V_2/V_1)	17 (D_2/D_1)	8 (V_{dc} std)
PMSG	inter-winding	6 (I_2/I_1)	88 (V_2/V_1)	4 (D_2/D_1)	3 (V_{dc} std)
SCIG	inter-turn	28 (I_2/I_1)	13 (V_2/V_1)	56 ($\text{PI}_d 2f_e$)	4 (V_{dc} std)
SCIG	inter-winding	33 (I_2/I_1)	0 (V_2/V_1)	67 ($\text{PI}_d 2f_e$)	0 (speed std)

matching impedance and all inter-winding trials (Table B1). Table 8 gives the physical quantity behind the relocation: under inter-winding faults the incremental negative-sequence impedance $\Delta V_2/\Delta I_2$ seen at the terminals is 15Ω in the PMSG, 7Ω in the SCIG and 11Ω in the WFSG, i.e. the PMSG drive converts a given negative-sequence current disturbance into twice the terminal voltage that the SCIG drive does; for inter-turn faults the PMSG’s negative-sequence current increments are of mixed sign and its median is indistinguishable from zero, which is why current-side features see little there. The two machines run the same field-oriented controller; what differs is the angle source (encoder against flux observer), the machine’s own negative-sequence impedance and the baseline of each feature, so the relocation is a property of the machine–drive combination rather than of the control structure alone.

5.3 Minimum detectable extent

Table 9 gives the minimum detectable extent (limit-of-detection convention) with its bootstrap interval, the detectable share with its Wilson interval and the median SDR with its tap-pair bootstrap interval, for four representative features; Fig. 4 plots the SDR against the design-level extent and Fig. 5 against the physical severity axis for all three alternatives. Through the terminal-voltage negative sequence the PMSG detects

Table 8 Healthy baselines of the three asymmetry ratios (median over fault trials, reference interval) and the incremental negative-sequence response under fault: median change of the negative-sequence current ΔI_2 (A, RMS) and terminal voltage ΔV_2 (V, RMS), and their ratio $\Delta V_2/\Delta I_2$ (Ω ; interquartile range in brackets), three-cycle windows; WFSG at the matching fault impedance.

Machine	Class	I_2/I_1	V_2/V_1	D_2/D_1	ΔI_2	ΔV_2	$\Delta V_2/\Delta I_2$
PMSG	inter-turn	0.0055	0.0009	0.0014	-0.000	-0.00	7.9 [-2.1, 16.5]
PMSG	inter-winding	0.0051	0.0009	0.0013	0.020	0.46	15.4 [10.9, 18.0]
SCIG	inter-turn	0.0079	0.0020	0.0036	0.001	0.01	6.7 [4.0, 7.9]
SCIG	inter-winding	0.0077	0.0020	0.0035	0.059	0.40	7.0 [6.6, 7.5]
WFSG	inter-turn	0.0148	0.0073	0.0039	0.003	-0.07	17.3 [11.8, 64.2]
WFSG	inter-winding	0.0148	0.0073	0.0038	0.027	0.30	10.6 [9.7, 12.8]

inter-turn faults from 3.8% [2.8, 3.8] and the SCIG from 7.4% [7.4, 7.4]; through the stator-current negative sequence both need 7.4%, and through the d -axis ripple the PMSG needs 12.1% and the SCIG 7.4% [3.8, 7.4]. The smallest tested inter-turn level (2.7–2.8%) is not reliably detectable on either machine by any single feature: the SCIG’s V_2/V_1 and PI ripple exceed one at 2.7% but not at 2.8 and 3.8%, which the limit-of-detection convention does not credit. For inter-winding faults the SCIG detects from the smallest tested extent (2.3%) on every feature except the q -axis voltage ripple, while the PMSG needs 2.3% [2.3, 14.3] on V_2/V_1 but 9.7% on I_2/I_1 and 21.7% on the d -axis ripple. The SDR rises with extent in every machine (Spearman 0.51–0.95), and the MDE intervals are wide because the tested extents form three clusters (about 3, 7.5 and 12% for inter-turn faults); the SCIG/PMSG ratio of median SDR is the better basis for comparison: 3.1 [1.3, 5.6] for I_2/I_1 and 12 [2.6, 30] for the d -axis ripple on inter-turn faults, but 1.1 [0.4, 4.6] for V_2/V_1 and 1.5 [0.5, 3.2] for the q -axis voltage ripple. The two machines differ where the signature relocates and are indistinguishable where it does not.

On the physical axis (Fig. 5) the picture is the same and independent of the tap-pair design: at equal fault-to-stator current ratio the SCIG’s current-side and d -axis features lie above the WFSG’s, which lie above the PMSG’s, while on V_2/V_1 the three machines nearly coincide. For the WFSG, whose small inter-turn extents were tested only at the lower impedance, the design-level MDE at the matching impedance is bounded by the smallest extent tested there (11.6%, detectable in 78–98% of trials depending on the suite); its 2.7–7.7% inter-turn trials at 1Ω sit at $\text{SDR} \approx 1$ on V_2/V_1 and I_2/I_1 , and all its inter-winding extents from 2.3% are detectable.

5.4 Operating envelope

Speed dominates the operating-point dependence (Table B3): in the PMSG the median SDR of the best feature and the detectable share rise from the lowest to the highest speed, in the SCIG less so and from a higher level; torque changes little. Two cautions apply. The dependence is consistent with a resistor-limited fault current that scales with speed (Section 5.1), so it is a property of the emulation as much as of the machines; and the pooled null is a mixture over operating points—the share of trials whose healthy change exceeds the pooled 95th percentile ranges from 0 to 16% by

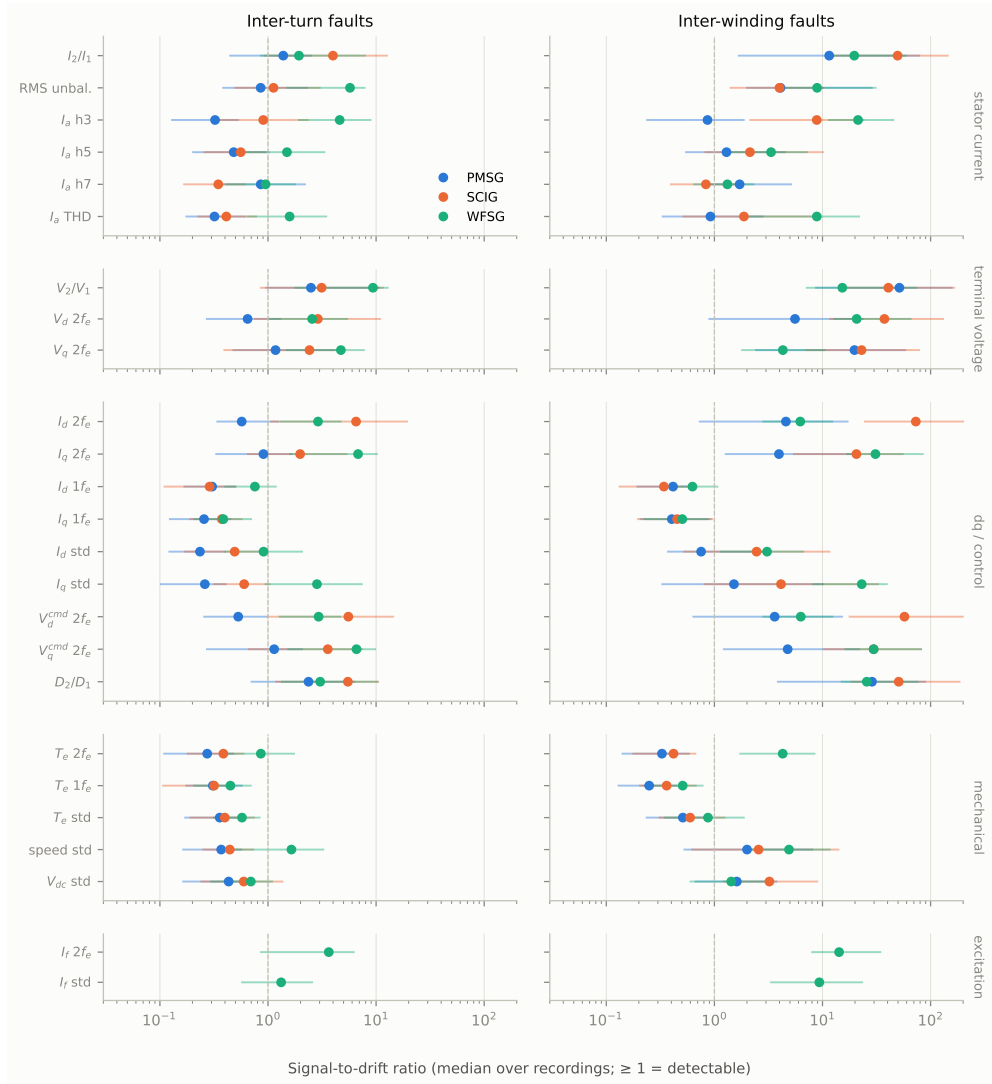


Fig. 3 Median signal-to-drift ratio of every feature for the three alternatives (dot) with interquartile range (bar), three-cycle windows; features to the right of the dashed line are detectable in more than half of the trials. The WFSG rows pool all its fault impedances.

speed for the key features—so the 5% false-alarm reading holds on average over the envelope, not at each point. Both make the lowest monitored speed a parameter of the requirement.

5.5 Observation suites and the design decision

Table 10 and Fig. 6 give the suite results for the PMSG and SCIG on five suites at five-cycle windows, Table B1 for the three alternatives on the common feature set at three

Table 9 Minimum detectable extent (MDE, % of the winding branch between the taps; limit-of-detection convention; bootstrap 95 % interval over operating points within each extent level), detectable share (DS, %; Wilson 95 % interval), median SDR (95 % interval from a bootstrap over tap pairs, the same pairs for both machines) and the SCIG/PMSG ratio of medians, for four representative features; five-cycle windows, $\alpha = 0.95$. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG. ‘-’: the largest tested extent is not detectable. Smallest tested extents: 2.7 % (inter-turn), 2.3 % (inter-winding).

Class	Feature	M.	MDE [CI]	DS [CI]	Median SDR [CI]	S/P ratio [CI]
IT	V_2/V_1	P	3.8 [2.8, 3.8]	76 [67, 83]	2.99 [1.21, 12.5]	1.1 [0.378, 4.56]
		S	7.4 [7.4, 7.4]	73 [64, 81]	4.17 [0.949, 15.5]	
IT	I_2/I_1	P	7.4 [7.4, 12.1]	56 [47, 65]	1.47 [0.601, 2.36]	3.07 [1.34, 5.63]
		S	7.4 [7.4, 7.4]	75 [66, 82]	4.52 [1.11, 11.7]	
IT	$PI_d 2f_e$	P	12.1 [11.6, 12.1]	31 [23, 40]	0.518 [0.389, 1.11]	11.8 [2.59, 28.8]
		S	7.4 [3.8, 7.4]	78 [69, 85]	6.69 [1.3, 18.3]	
IT	$V_q 2f_e$	P	7.4 [7.4, 11.6]	57 [48, 66]	1.35 [0.791, 4.58]	1.49 [0.464, 3.21]
		S	7.4 [7.4, 7.7]	60 [51, 69]	2.58 [0.516, 5.33]	
IW	V_2/V_1	P	2.3 [2.3, 14.3]	94 [88, 97]	61.9 [9.24, 181]	0.974 [0.696, 1.68]
		S	2.3 [2.3, 2.3]	100 [97, 100]	48.2 [15.3, 188]	
IW	I_2/I_1	P	9.7 [9.5, 21.7]	81 [73, 88]	11.3 [1.98, 64.3]	5.49 [1.75, 13.1]
		S	2.3 [2.3, 2.3]	100 [97, 100]	55.6 [12.5, 117]	
IW	$PI_d 2f_e$	P	21.7 [9.7, 21.7]	72 [63, 80]	4.21 [1.18, 13]	18.2 [12.3, 27.2]
		S	2.3 [2.3, 2.3]	100 [97, 100]	74 [22.9, 245]	
IW	$V_q 2f_e$	P	9.7 [2.3, 14.3]	90 [83, 94]	22.4 [3.24, 62]	1.13 [0.831, 1.9]
		S	9.7 [2.3, 9.7]	92 [85, 96]	24.8 [7.04, 78.6]	

cycles, and Table 11 the decision table of step 5. For the PMSG, current transducers alone resolve inter-turn faults from 7.4 % and inter-winding faults from 9.7 %; adding three voltage transducers takes these to 3.8 and 2.3 %, and neither controller access nor a torque transducer improves the fixed-feature result further (controller access raises the oracle share from 88 to 95 %). The drive-internal suite alone reaches 7.4 % for inter-turn faults through the duty-cycle negative sequence, which carries a weaker version of the signature that the measured terminal voltage carries. For the SCIG the cheapest suites already reach what any suite reaches—7.4 % for inter-turn faults through I_2/I_1 or, with no added sensor, the d -axis PI output, and 2.3 % for inter-winding faults—and additional hardware raises the detectable share rather than the reach. For the WFSG every suite detects all inter-winding faults from 2.3 % and 78–98 % of the inter-turn trials at the matching impedance, and the field current alone detects 86 and 100 %. The nested selection of the fixed feature changes no share by more than five points (Table B1), so the in-sample choice is not what drives these numbers. The oracle shares, however, come with false-alarm rates of 44–78 % on the healthy trials for the suites with many features (Table B1): the per-trial best of twenty features is not a calibrated detector, and the oracle column is an upper bound only. A shaft torque transducer adds nothing for winding faults on the two machines on which it was available (the torque features’ healthy relative change exceeds 100 % on the SCIG and their median SDR is below 0.3 on the PMSG).

The decision table turns this into the selection of step 5. At $\alpha = 0.95$ no suite on any alternative meets an inter-turn requirement of 3 or 5 % with 95 % bootstrap confidence; the PMSG’s voltage suite is the only candidate below 5 % and its verdict

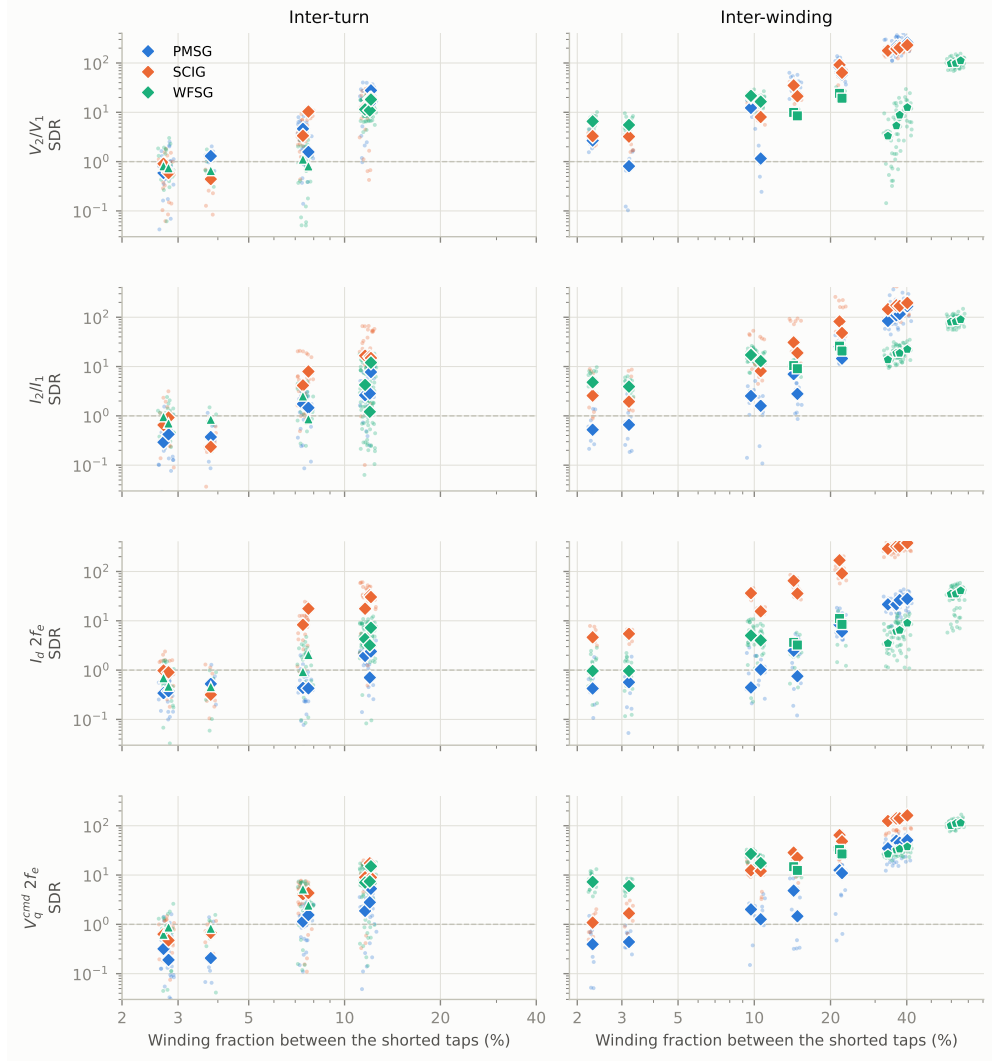


Fig. 4 Signal-to-drift ratio against the design-level fault extent for four features and the three alternatives (small dots: trials; large markers: median per extent level and fault impedance, marker shape encoding the impedance), three-cycle windows.

is uncertain, a consequence of the three-cycle windows and the coarse extent grid. At 8% the drive-internal suite meets the requirement on the PMSG and the SCIG, i.e. no added sensor is needed for either; the WFSG's inter-turn verdicts are bounded by its smallest tested extent. For inter-winding faults the SCIG and the WFSG meet every requirement from 3% with the drive-internal suite, whereas the PMSG needs current transducers at 12% and is uncertain below (its five-cycle result, Table 10, reaches 2.3% with voltage transducers). Relaxing α to 0.90 admits the PMSG's voltage suite at 5% for inter-turn faults; tightening it to 0.99 removes the PMSG's current-only

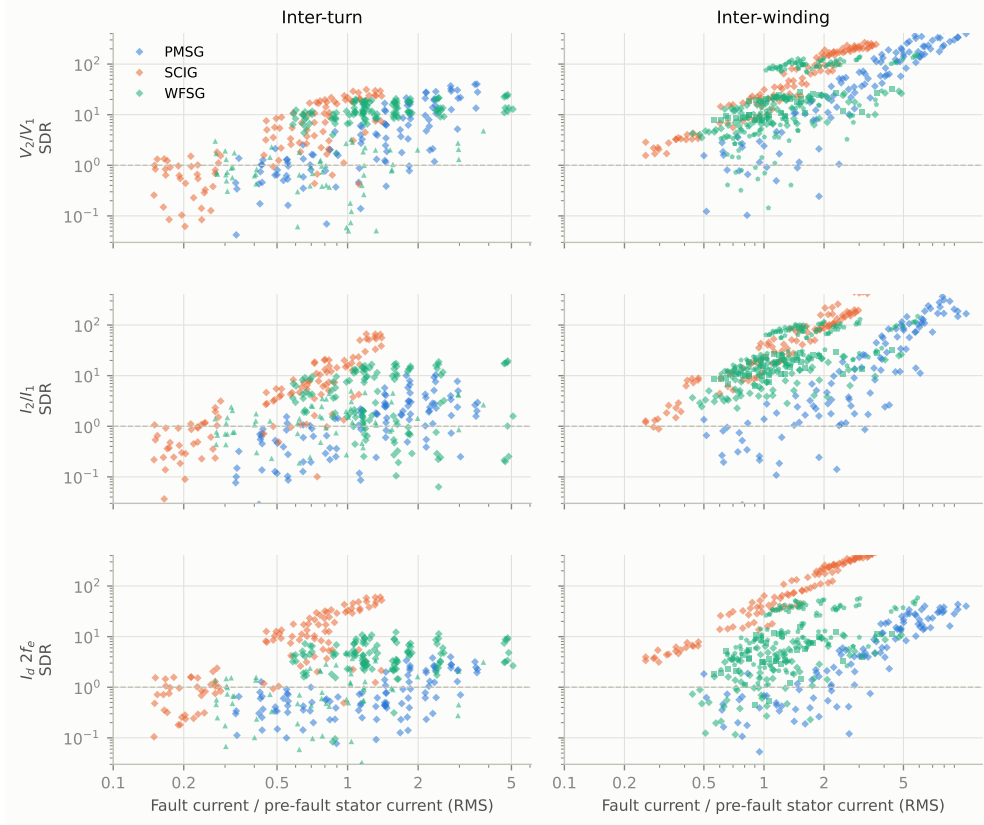


Fig. 5 Signal-to-drift ratio against physical fault severity (RMS fault current relative to the RMS pre-fault stator current) for three features and the three alternatives, all trials, three-cycle windows. Marker shape encodes the fault impedance of the WFSG trials.

option at 12 % for inter-winding faults. The design content is therefore twofold: for a requirement around 8 % of a branch, the three alternatives are equally diagnosable with drive-internal quantities and diagnosability does not discriminate between them; for a requirement below 5 %, only the PMSG with voltage transducers is a candidate and the evidence is not yet conclusive. In neither case would the architecture decision change on diagnosability grounds; the monitoring specification would.

5.6 Transfer across alternatives

Table 12 reports step 6 for the PMSG–SCIG pair and Table 13 for the three alternatives; the metrics on all windows and on the out-of-reference windows agree to within 0.02, so the windows used for normalisation do not drive the result. In physical units a gradient-boosted detector reaches an AUC of 0.87–0.91 within a machine on unseen tap pairs but 0.78 (PMSG→SCIG) and 0.55 (SCIG→PMSG) across; with its features scaled on the target machine’s healthy trials, a commissioning baseline, 0.82 and 0.75, and a logistic detector under that rule falls below chance in one direction (0.45); with

Table 10 Observation suites versus what they detect, PMSG and SCIG (five-cycle windows): best fixed feature of the suite, its minimum detectable extent (MDE, %; limit-of-detection convention) and detectable share (DS, %; per-trial oracle share in brackets). Features failing the reliability screen (three or more false alarms on the nine healthy trials, or null quantile above one) are excluded for that machine; the oracle false-alarm rates are in Table B1.

Suite	PMSG feature	MDE	DS [oracle]	SCIG feature	MDE	DS [oracle]
<i>Inter-turn faults</i>						
drive-internal	D_2/D_1	7.4	64 [87]	$PI_d 2f_e$	7.4	78 [99]
3 CT	I_2/I_1	7.4	56 [77]	I_2/I_1	7.4	75 [87]
3 CT + 3 VT	V_2/V_1	3.8	76 [88]	I_2/I_1	7.4	75 [90]
3 CT + 3 VT + drive	V_2/V_1	3.8	76 [95]	$PI_d 2f_e$	7.4	78 [99]
+ torque transducer	V_2/V_1	3.8	76 [95]	$PI_d 2f_e$	7.4	78 [99]
<i>Inter-winding faults</i>						
drive-internal	D_2/D_1	14.3	90 [100]	$PI_d 2f_e$	2.3	100 [100]
3 CT	I_2/I_1	9.7	81 [94]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT	V_2/V_1	2.3	94 [97]	I_2/I_1	2.3	100 [100]
3 CT + 3 VT + drive	V_2/V_1	2.3	94 [100]	$PI_d 2f_e$	2.3	100 [100]
+ torque transducer	V_2/V_1	2.3	94 [100]	$PI_d 2f_e$	2.3	100 [100]

Table 11 Decision table: cheapest observation suite whose minimum detectable extent meets the requirement e_{req} with bootstrap probability ≥ 0.95 , per alternative, fault class and false-alarm level α (three-cycle windows; suites ordered drive-internal < 3CT < 3CT+3VT < 3CT+3VT+drive; exc: WFSG field current). ‘(suite)?’: no suite meets with that confidence, the cheapest suite whose verdict is uncertain is shown; ‘none’: every suite fails. WFSG inter-turn verdicts are bounded by its smallest tested extent at the matching impedance (11.6%).

α	e_{req} (%)	PMSG		SCIG		WFSG	
		IT	IW	IT	IW	IT	IW
0.90	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.90	5	3CT+3VT	(drive)?	none	drive	none	drive
0.90	8	drive	(drive)?	drive	drive	none	drive
0.90	12	drive	3CT	drive	drive	drive	drive
0.95	3	(3CT+3VT)?	(drive)?	none	drive	none	drive
0.95	5	(drive)?	(drive)?	none	drive	none	drive
0.95	8	drive	(drive)?	drive	drive	none	drive
0.95	12	drive	3CT	drive	drive	drive	drive
0.99	3	none	none	none	drive	none	drive
0.99	5	(3CT+3VT)?	none	none	drive	none	drive
0.99	8	drive	none	drive	drive	none	drive
0.99	12	drive	(drive)?	drive	drive	drive	drive

each feature referenced to the first half of the trial’s own reference interval, 0.91 and 0.84, against 0.91–0.94 within a machine, and a logistic detector transfers at 0.89 and 0.87. The threshold that gives 1% false positives on all negatives fires on 0–3% of the healthy trials’ fault-time windows, the only negatives that contain the contactor operation. Between the three alternatives, self-referenced detectors transfer in all six directions with AUC 0.80–0.98 on the out-of-reference windows, and a detector trained

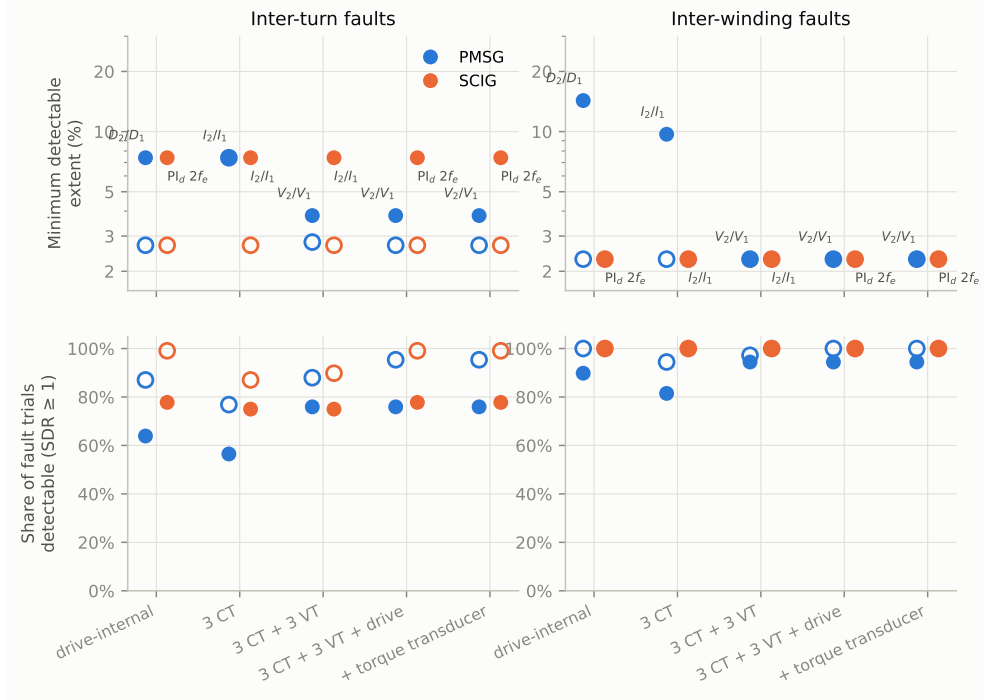


Fig. 6 Observation suites versus what they detect, PMSG and SCIG (five-cycle windows): minimum detectable extent (top; the fixed feature is named) and detectable share (bottom) for the fixed best feature of each suite (filled) and the per-trial oracle (hollow).

Table 12 Transfer of a window-level detector between PMSG and SCIG: AUC on all windows / AUC on the windows outside the normalising reference windows (second half of the reference interval, recovery windows and fault-time windows of the healthy trials). Within-machine columns use five-fold cross-validation grouped by tap pair (healthy trials grouped by speed); cross columns train on all trials of one machine and test on all trials of the other.

Referencing	Model	P→P	S→S	P→S	S→P
physical units	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.59/0.59
physical units	GBDT	0.88/0.87	0.90/0.91	0.78/0.78	0.55/0.55
z on target's healthy trials	logistic	0.78/0.78	0.79/0.80	0.71/0.71	0.44/0.45
z on target's healthy trials	GBDT	0.88/0.87	0.90/0.91	0.82/0.82	0.75/0.75
z vs own reference	logistic	0.90/0.89	0.94/0.93	0.90/0.89	0.88/0.87
z vs own reference	GBDT	0.92/0.91	0.95/0.94	0.92/0.91	0.86/0.84

on two machines detects the faults of the third with 0.83 (PMSG), 0.88 (SCIG) and 0.97 (WFSG); the WFSG entries are optimistic because that bench has no healthy trials and therefore no nuisance-transient negatives. Restricted to each suite's features (Table B6), every suite transfers at 0.82–0.95 when self-referenced and at 0.62–0.90 under the commissioning baseline.

Table 13 Transfer between the three alternatives (three-cycle windows, common features, $|z|$ against the first half of each trial’s own reference interval, GBDT with fixed seed). AUC on all windows and on the out-of-reference windows; TPR at 1 % FPR on the out-of-reference windows; FA: share of the fault-time windows of the healthy trials (contactor operation without a fault) above the threshold that gives 1 % FPR on all negatives. Same-machine rows: five-fold cross-validation grouped by tap pair. The WFSG has no healthy trials, so its negatives are reference and recovery windows only and its rows are optimistic.

Trained on	Tested on	AUC all	AUC o.o.r.	TPR o.o.r.	FA healthy (%)
PMSG	PMSG	0.86	0.85	0.62	3
SCIG	SCIG	0.92	0.91	0.76	1
WFSG	WFSG	0.97	0.97	0.93	–
PMSG	SCIG	0.88	0.88	0.60	0
PMSG	WFSG	0.97	0.98	0.89	–
SCIG	PMSG	0.81	0.80	0.35	0
SCIG	WFSG	0.95	0.96	0.86	–
WFSG	PMSG	0.86	0.84	0.52	1
WFSG	SCIG	0.91	0.89	0.61	0
SCIG+WFSG	PMSG	0.84	0.83	0.41	0
PMSG+WFSG	SCIG	0.89	0.88	0.61	0
PMSG+SCIG	WFSG	0.97	0.97	0.88	–

5.7 Robustness and the two nulls

Table B4 shows how the tuning choices move the design quantities, with the screen applied at every setting. The inter-turn MDEs of all five key features are the same at three, five and eight cycles on both machines, and the detectable shares move by at most six points; for inter-winding faults one MDE moves (the PMSG’s V_2/V_1 , 9.7 % at three cycles against 2.3 % at five and eight). Tightening the null quantile from 0.95 to 0.99 lowers the shares by 3–9 points and moves two PMSG inter-winding MDEs up one level; relaxing it to 0.90 raises the shares by up to 9 points and moves one down. No ordering of the two machines on any feature reverses, but the suite conclusion for the PMSG’s inter-winding faults does depend on α and on the window length, as the decision table shows. The screen removes between one and five features per machine and setting (Table B5); none of the five key features is removed at any setting, and the 3-cycle result of V_2/V_1 on the PMSG (two false alarms in nine healthy trials) passes the binomial rule. An absolute-change variant of the SDR, which removes the healthy level of the feature from both numerator and null, ranks the 30 features with a Spearman correlation of 0.92–0.98 against the relative variant and picks the same three top features in every machine and class. The bootstrap intervals (Table 9, Fig. 7) show the tap-pair-level variability behind each number: the median SDRs are known to within a factor of two to four, the shares to ± 8 –10 points, and the ratios of median SDR exclude one where the signature relocates and include it where it does not.

Table 14 quantifies the assumption that the healthy reference lies immediately before the fault. When the reference is instead the healthy trial recorded at the same operating point at another time—a commissioning baseline—the null widens by a factor of two to seven, the median SDR of the inter-turn features falls to about one (PMSG V_2/V_1 from 3.0 to 1.1; SCIG I_2/I_1 from 4.5 to 1.1) and their detectable shares to 44–59 %, while inter-winding faults remain detectable in 67–94 % of trials. The

Table 14 Within-trial null (reference immediately before the fault) versus commissioning-baseline null (reference: the healthy trial at the same operating point, recorded at a different time): null quantile q_{95} , median SDR and detectable share DS (%), five-cycle windows, $\alpha = 0.95$. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

Class	Feature	M.	q_{95}		Median SDR		DS	
			within	baseline	within	baseline	within	baseline
IT	V_2/V_1	P	0.086	0.315	2.99	1.05	76	50
IT	I_2/I_1	P	0.162	0.268	1.47	0.978	56	48
IT	$I_d 2f_e$	P	0.563	0.876	0.516	0.441	30	20
IT	$PI_d 2f_e$	P	0.557	0.870	0.518	0.441	31	20
IT	$V_q 2f_e$	P	0.165	0.277	1.35	0.872	57	44
IT	D_2/D_1	P	0.090	0.211	2.78	1.11	64	52
IW	V_2/V_1	P	0.086	0.315	61.9	12.5	94	83
IW	I_2/I_1	P	0.162	0.268	11.3	6.34	81	75
IW	$I_d 2f_e$	P	0.563	0.876	4.14	2.4	72	67
IW	$PI_d 2f_e$	P	0.557	0.870	4.21	2.4	72	67
IW	$V_q 2f_e$	P	0.165	0.277	22.4	10.7	90	81
IW	D_2/D_1	P	0.090	0.211	32.6	11.9	90	85
IT	V_2/V_1	S	0.033	0.111	4.17	1.45	73	59
IT	I_2/I_1	S	0.034	0.130	4.52	1.15	75	54
IT	$I_d 2f_e$	S	0.016	0.078	6.68	1.29	79	53
IT	$PI_d 2f_e$	S	0.016	0.078	6.69	1.29	78	53
IT	$V_q 2f_e$	S	0.060	0.131	2.58	1.54	60	58
IT	D_2/D_1	S	0.010	0.068	11.1	1.57	81	52
IW	V_2/V_1	S	0.033	0.111	48.2	16.1	100	94
IW	I_2/I_1	S	0.034	0.130	55.6	12.9	100	86
IW	$I_d 2f_e$	S	0.016	0.078	74	16	100	88
IW	$PI_d 2f_e$	S	0.016	0.078	74	16	100	88
IW	$V_q 2f_e$	S	0.060	0.131	24.8	11.2	92	86
IW	D_2/D_1	S	0.010	0.068	84.9	12.2	100	82

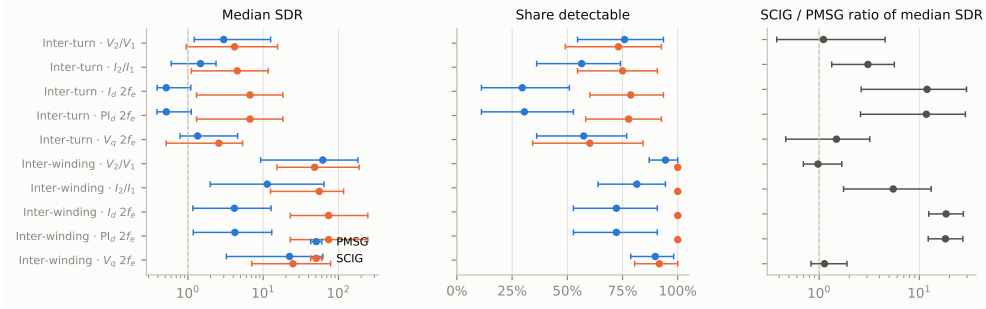


Fig. 7 Tap-pair-level uncertainty: bootstrap 95 % intervals of the median SDR, the detectable share and the SCIG/PMSG ratio of median SDR for five features and both fault classes (the same resampled tap pairs for both machines).

MDEs reported in this paper are therefore statements about a short-horizon change detector; a monitor that compares against an old baseline needs the wider null, and the transfer result of Section 5.6 says the same thing from the other side.

6 Discussion

6.1 What the demonstration establishes about the method

The method was proposed as a way of making diagnosability a criterion in architecture selection. The demonstration supports four claims about it that do not depend on which machine came out ahead, and it corrects the naive form of two of them.

The design-level description of a failure mode does not fix its physical severity. A short-circuit of the same winding fraction produced a fault current that differed by about 10% between the PMSG and the SCIG in amperes and by a factor of 2.3 relative to the stator current the sensors see (Section 5.1); on the WFSG bench the extent and the fault impedance were confounded by the experimenters. Any method that compares alternatives on a failure-mode class must therefore carry two severity axes—the design-level one, on which requirements are written, and a physical one, on which the alternatives are compared—and must state which one a number refers to. For resistor-limited emulation the physical axis is predictable at the design stage from volts per turn, speed and the resistance; for a bolted fault it would be predictable from the leakage reactance of the shorted turns, and the topologies would differ further in whether the fault can be de-excited.

Where a failure becomes observable is a property of the machine-drive combination, and the SDR must be read with its decomposition. In the PMSG the winding asymmetry appeared in the terminal voltage, in the SCIG in the magnetising-axis current and its controller action, in the WFSG additionally in the field current and the current harmonics; the incremental negative-sequence impedance under fault was twice as large in the PMSG drive as in the SCIG drive (Table 8). The version of this claim in which “the control structure decides” is not supported: both machines run the same controller, and the decomposition of Table 6 shows that the SCIG’s higher SDR on current and d -axis features is entirely a quieter healthy null, its relative changes under fault being smaller than the PMSG’s. The SDR is designed to credit a quiet alternative—a fault is detectable relative to what the healthy system does on its own—but a design team must know whether an alternative wins by signature or by silence, because the two generalise differently: a signature is a property of the fault physics, a null of the sensing and control chain and of the operating regime, which is exactly what a design-stage instantiation cannot simulate (Section 3.5).

The cheapest adequate observation suite depends on the alternative and on the requirement, and in this demonstration it would change the monitoring specification, not the architecture. With three current transducers the PMSG reaches 7.4% (inter-turn) and 9.7% (inter-winding); three voltage transducers take it to 3.8 and 2.3% and buy nothing on the SCIG, whose current-only or drive-internal suites already reach 7.4 and 2.3% (Table 10). The decision table (Table 11) makes the requirement explicit: for a required extent around 8% of a branch all three alternatives meet it with drive-internal quantities, and diagnosability does not discriminate between them; below 5% only the PMSG with voltage transducers is a candidate, with an uncertain verdict at the confidence demanded. The method would change an architecture decision only if, for one alternative, no affordable suite met the requirement; here it changes what is specified for each. That is the honest answer to “so what for the designer”: a measured

detection rating per alternative and suite, with its uncertainty, in place of an assigned one.

A monitoring design transfers across alternatives only under the right referencing rule, and the rule is a short-horizon one. A detector built on one machine and applied to another in physical units lost much of its performance; features scaled on the target machine’s healthy trials—a commissioning baseline—recovered part of it and were harmful in one direction; features referenced to the first half of each trial’s own reference interval transferred in all six directions between the three machines and from any two to the third (Section 5.6). The between-trial null of Table 14 explains why: against a baseline recorded at another time, the inter-turn signatures sit at $\text{SDR} \approx 1$. For a product family this is a specification rule: the monitoring function should be defined as a change detector on self-referenced deviations over a short horizon, and a monitor that compares against an old baseline must be calibrated on the wider null.

6.2 Design implications

For the wind-turbine generator, the results translate into statements a design team can use, with the scope of Section 6.4. A PMSG with full-power conversion should be specified with terminal voltage measurement—or another observable of the terminal asymmetry, such as the zero-sequence voltage where a neutral is accessible (Urresty et al., 2012)—if inter-turn faults below about 7% of a branch are to be detected by threshold monitoring; its controller’s commanded quantities carry a weaker and less reliable version of the signature. An SCIG reaches the same detectability with current transducers alone or with no added sensor at all, and additional hardware buys robustness rather than reach. A wound-field machine with a static exciter offers a further drive-internal channel, the field current, whose $2f_e$ ripple detected every inter-winding fault and most inter-turn faults; a brushless exciter would not expose it (Neti and Nandi, 2009). A shaft torque transducer, as filtered on these benches, is not worth specifying for winding faults. A monitoring function meant to serve more than one generator type should be specified on self-referenced deviations. Two industrial parameters change these statements and are computable at the concept stage: the ratio of current-loop bandwidth to twice the electrical frequency, which for a direct-drive multi-megawatt PMSG at a fundamental near 10 Hz will push the signature further into the controller and the voltage than in these 40–60 Hz machines; and the presence of a grid-connected, unregulated stator in a doubly-fed generator, where the migration is absent and current-signature routes regain their reach.

More generally, the method gives DFMEA’s detection rating a measured basis: the “D” of a failure mode becomes a function of the alternative and the observation suite, with an uncertainty interval, rather than a consensus value. It also gives digital-twin efforts a concrete design-stage use with a clear division of labour: the twin supplies the faulted numerator for each alternative, measured healthy records of hardware of the same class supply the null, and a few physical trials validate the pair.

6.3 Applying the method elsewhere

Nothing in Section 3 is specific to electrical machines, but the scope conditions of Section 3.6 are. A gearbox family would use tooth-damage size as extent, vibration and oil-debris suites, and seeded-damage trials with a healthy run immediately before each damage step; a battery-pack family would use cell-capacity loss, voltage, temperature and impedance suites, and trials in which a degraded cell is switched into a healthy pack; a hydraulic actuator would use leakage cross-section, pressure, flow and position suites, and bench trials. Run-to-failure records do not fit the trial structure without a different null. The SDR, the pooled null, the binomial screen, the fixed, nested and oracle variants, the decision table and the transfer test carry over unchanged.

6.4 Threats to validity

The most important limitation is that each alternative is represented by one physical machine. The differences reported are differences between three machines; attributing them to topology rather than to the individual units requires replicated machines or simulated populations of each alternative under parameter variation, validated against the measured units. We regard the demonstration as establishing the method and the mechanisms, not the population-level ranking of the three topologies, and the wording of the findings is scaled accordingly. The two benches differ in converter, controller, sampling rate and sensing chain; the WFSG's healthy null differs from the SCIG's although its measured features are similar, and the decomposition shows how much of any SDR difference is null.

The fault emulation is resistor-limited and the benches are laboratory scale (2–2.5 kVA); the operating points were scaled from a 15 MW turbine but the electrical frequency and the fault physics were not. Only winding faults and only electrical and controller observables were studied, and the suites are those the datasets can populate: zero-sequence voltage, stray flux, temperature, vibration and partial discharge are absent, and the angle-free Park's-vector features were not computed. On the WFSG the fault impedance is confounded with extent; only four tap pairs per class exist at the impedance matching the other two machines, so its inter-turn MDE is bounded, not measured; it has no healthy trials, so the reliability screen cannot be applied to it and its transfer entries lack nuisance-transient negatives; and its fault intervals are short, so that 81 of its 438 trials hold fewer than three three-cycle windows and four hold none. The PMSG/SCIG faulted interval starts at the contactor command and therefore contains about 55 ms of healthy time before the relay picks up, which dilutes those SDRs slightly relative to the WFSG's, whose interval is located from the measured fault current.

Statistically, the pooled null is a mixture over operating points and holds its nominal false-alarm rate on average, not per speed; the null quantile is held fixed inside the bootstrap; the fixed feature is chosen on the trials it is evaluated on, which the nested variant shows to be immaterial here; the oracle variant is a maximum over features and its false-alarm rate is reported for that reason; the minimum detectable extent is a discrete quantity on a coarse extent grid, and its intervals are correspondingly wide, so the continuous SDR and its ratio are the better basis for comparing

alternatives. The transfer experiment’s true-positive rate at 1 % false-positive rate is a point of the receiver operating characteristic rather than an operating threshold, and other nuisance transients (load steps, grid events) were not in the data. Finally, the SDR characterises a short-horizon change detector; the between-trial null shows that a monitor referenced to an old baseline faces a null two to seven times wider, and every MDE in this paper should be read with that qualification.

7 Conclusions

We have proposed a method that makes the diagnosability of failure modes a criterion for choosing between architectural alternatives and, jointly, for choosing the observation suite that will monitor the chosen architecture. Its core is a scale-free, self-calibrated statistic, the signal-to-drift ratio, from which a minimum detectable extent and a detectable share above a required extent can be computed for every combination of alternative, suite and failure-mode class, with a reliability screen, bootstrap intervals and a decision table over the required extent and the false-alarm level; a six-step procedure with default rules for its framing decisions turns these into a co-selection and a transferability specification.

Applied to three wind-turbine generator topologies subjected to identical winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault across the machines; that where a fault becomes visible depends on the machine–drive combination, with the difference in detectability between the two machines on one bench arising from their healthy non-stationarity rather than from the strength of the signature; that the cheapest observation suite meeting a requirement depends on the alternative and on the requirement, terminal voltage transducers lowering the permanent-magnet machine’s detectable extent and adding nothing for the induction machine, while for a requirement of about 8 % of a branch all three alternatives are equally diagnosable with drive-internal quantities; and that a learned detector transfers across the three machines only when its features are referenced to each trial’s own preceding healthy behaviour, a commissioning baseline being insufficient for inter-turn faults. The findings are robust in their orderings to the method’s tuning, are reported with tap-pair-level uncertainty, and are limited by one physical unit per alternative, laboratory scale, resistor-limited fault emulation and a single failure-mode class.

Three extensions follow directly. First, the design-stage instantiation: simulated faulted trials of each alternative combined with measured healthy nulls of hardware of the same class, validated on a few physical trials, would let the method be applied before hardware exists and would turn the demonstration’s machine-level differences into population-level statements. Second, other failure-mode classes and observables—bearing and rotor faults with vibration and flux observables, zero-sequence and Park’s-vector features for the winding faults studied here—would test the method’s claim to generality within the domain, and a mechanical or electrochemical family would test it across domains. Third, the cost side of the selection, treated here as an ordinal ranking of suites, deserves a model that includes integration, maintenance and the cost of missed detection, so that the co-selection can be posed as an optimisation.

The code released with this paper reproduces every result from the public data and is intended as the starting point for those extensions.

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Declarations

Funding. No specific funding was received for this work.

Competing interests. The author declares no competing interests.

Ethics approval and consent. Not applicable; the work uses no human or animal subjects and no personal data.

Data availability. All data are public: the PMSG, SCIG and synchronous-generator datasets of the InnovaPower MitDev-Eletrica project (<https://github.com/InnovaPower/MitDev-Eletrica>; Zenodo DOIs 10.5281/zenodo.15741561, 10.5281/zenodo.17161986 and 10.5281/zenodo.13685630), licensed CC BY 4.0. The derived per-trial tables (features, signal-to-drift ratios, bootstrap samples) are provided with the code.

Code availability. The complete analysis pipeline (feature extraction, signal-to-drift computation, sensor-suite evaluation, transfer experiments, sensitivity analysis and all figures) is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`; an archived release with a DOI will be deposited on Zenodo at acceptance); it reproduces every number in this paper from the public data.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction as an assistant for code development, literature retrieval, data analysis and drafting and editing of the text. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A Feature definitions

Let x_a, x_b, x_c be the sampled phase quantities of a window of N samples at sampling frequency f_s , w the Hann window and f_e the electrical frequency of the window. The complex fundamental amplitude of a signal x at frequency f is

$$X(f) = \frac{2}{\sum_n w_n} \sum_{n=0}^{N-1} x_n w_n e^{-j2\pi f n / f_s}. \quad (\text{A1})$$

The electrical frequency is estimated from the slope of the unwrapped phase of the current space vector $\frac{2}{3}(i_a + \mathbf{a} i_b + \mathbf{a}^2 i_c)$, $\mathbf{a} = e^{j2\pi/3}$, whose sign gives the phase rotation.

Sequence magnitudes are $|X_+| = |X_a + \mathbf{a}X_b + \mathbf{a}^2X_c|/3$ and $|X_-| = |X_a + \mathbf{a}^2X_b + \mathbf{a}X_c|/3$ for an abc rotation (interchanged for acb), evaluated at $f = f_e$ for currents (I_1, I_2), terminal voltages (V_1, V_2) and mean-removed duty cycles (D_1, D_2); $|X(f)|$ is a fundamental amplitude (peak), and is divided by $\sqrt{2}$ wherever it is compared with an RMS value. The RMS unbalance is $\max_k |I_{\text{rms},k} - \bar{I}_{\text{rms}}|/\bar{I}_{\text{rms}}$ over the three phases. Harmonic ratios are $|I_a(kf_e)|/|I_a(f_e)|$ for $k = 3, 5, 7$, and the THD is $\sqrt{\sum_{k=2}^{20} |I_a(kf_e)|^2}/|I_a(f_e)|$. For the synchronous-frame currents i_d, i_q and the torques the features are the standard deviation of the mean-removed signal and its amplitudes $|Y(f_e)|$ and $|Y(2f_e)|$; for the PI outputs the standard deviation and $|Y(2f_e)|$; for the measured and commanded synchronous-frame voltages $|Y(2f_e)|$ only; for the field current the standard deviation and $|Y(2f_e)|$. Speed and DC-link features are standard deviations. Windows are n_c electrical cycles long with a hop of one cycle; a feature is undefined for a trial whose faulted interval holds no complete window.

Appendix B Supplementary tables

This appendix collects the three-alternative suite table, the per-alternative feature ranking, the operating-point grid, the sensitivity of the design quantities to the tuning choices, the outcome of the reliability screen and the transfer of suite-restricted detectors.

Table B1 Observation suites for the three alternatives, three-cycle windows, common feature set, $\alpha = 0.95$. Fixed feature: highest median SDR among the suite’s screened features. MDE (%) at the fault impedance matching the PMSG/SCIG protocol (WFSG: 198 trials, four tap pairs per class); † the smallest inter-turn extent tested on the WFSG at that impedance is 11.6%, so the value is an upper bound. DS: detectable share (%) of the fixed feature; nested: fixed feature chosen on the other tap pairs; oracle: per-trial best feature, with its false-alarm rate on the healthy trials (FA, %; not available for the WFSG). ‘-’: largest tested extent not detectable.

Suite	Machine	Feature	MDE	DS	DS nested	Oracle DS	Oracle FA
<i>Inter-turn faults</i>							
drive-internal	PMSG	D_2/D_1	7.4	65	65	82	44
drive-internal	SCIG	$I_d 2f_e$	7.4	78	78	93	78
drive-internal	WFSG	$I_q 2f_e$	11.6 †	91	87	96	–
3 CT	PMSG	I_2/I_1	7.4	57	57	78	11
3 CT	SCIG	I_2/I_1	7.4	70	70	86	56
3 CT	WFSG	$I_a h3$	11.6 †	98	86	94	–
3 CT + 3 VT	PMSG	V_2/V_1	3.8	73	73	85	33
3 CT + 3 VT	SCIG	I_2/I_1	7.4	70	70	89	56
3 CT + 3 VT	WFSG	V_2/V_1	11.6 †	98	98	94	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	3.8	73	68	94	67
3 CT + 3 VT + drive	SCIG	$I_d 2f_e$	7.4	78	78	97	89
3 CT + 3 VT + drive	WFSG	V_2/V_1	11.6 †	98	98	98	–
excitation (field current)	WFSG	$I_f 2f_e$	11.6 †	86	86	74	–
<i>Inter-winding faults</i>							
drive-internal	PMSG	D_2/D_1	14.3	91	91	98	44
drive-internal	SCIG	$I_d 2f_e$	2.3	100	100	100	78
drive-internal	WFSG	$I_q 2f_e$	2.3	100	100	100	–
3 CT	PMSG	I_2/I_1	9.7	83	83	99	11
3 CT	SCIG	I_2/I_1	2.3	98	98	99	56
3 CT	WFSG	I_2/I_1	2.3	100	100	100	–
3 CT + 3 VT	PMSG	V_2/V_1	9.7	92	92	99	33
3 CT + 3 VT	SCIG	I_2/I_1	2.3	98	98	100	56
3 CT + 3 VT	WFSG	V_2/V_1	2.3	100	100	100	–
3 CT + 3 VT + drive	PMSG	V_2/V_1	9.7	92	92	100	67
3 CT + 3 VT + drive	SCIG	$I_d 2f_e$	2.3	100	100	100	89
3 CT + 3 VT + drive	WFSG	$I_q 2f_e$	2.3	100	100	100	–
excitation (field current)	WFSG	$I_f 2f_e$	2.3	100	100	100	–

Table B2 Five features with the highest median SDR per alternative and fault class (three-cycle windows; WFSG at all its fault impedances, trials with at least one full fault window); detectable share in parentheses (%). †: fails the reliability screen for that machine (three or more false alarms on the nine healthy trials, or null quantile above one) and is therefore not used in the suite tables.

Machine	Class	Features: median SDR (DS)
PMSG	inter-turn	V_2/V_1 2.5 (73); D_2/D_1 2.4 (65); I_2/I_1 1.4 (57); $V_q 2f_e$ 1.2 (55); $V_q^{\text{cmd}} 2f_e$ 1.1 (55)
PMSG	inter-winding	V_2/V_1 51.5 (92); D_2/D_1 28.7 (91); $V_q 2f_e$ 19.8 (89); I_2/I_1 11.5 (83); $V_d 2f_e$ 5.6 (71)
SCIG	inter-turn	$I_d 2f_e$ 6.5 (78); $V_d^{\text{cmd}} 2f_e$ 5.5 (73); D_2/D_1 5.5 (78); I_2/I_1 4.0 (70); $V_q^{\text{cmd}} 2f_e$ 3.6 (66)
SCIG	inter-winding	$I_d 2f_e$ 73.0 (100); $V_d^{\text{cmd}} 2f_e$ 57.3 (100); D_2/D_1 50.7 (100); I_2/I_1 49.5 (98); V_2/V_1 40.7 (100)
WFSG	inter-turn	V_2/V_1 9.4 (80); $I_q 2f_e$ 6.8 (81); $V_q^{\text{cmd}} 2f_e$ 6.6 (80); RMS unbal. 5.7 (80); $V_q 2f_e$ 4.7 (81)
WFSG	inter-winding	$I_q 2f_e$ 30.9 (100); $V_q^{\text{cmd}} 2f_e$ 29.7 (100); D_2/D_1 25.7 (100); I_q std 23.1 (95); I_a h3 21.3 (100)

Table B3 Operating-point dependence: median SDR of the best of V_2/V_1 , $V_q 2f_e$, I_2/I_1 , $PI_d 2f_e$ and detectable share (%) per cell of the speed–torque grid, 24 fault trials per cell.

Machine	Speed (rpm)	5.2 N m	6.4 N m	8.0 N m
PMSG	1200	7.9 / 83	9.0 / 83	5.3 / 79
PMSG	1500	12.8 / 96	11.3 / 96	9.6 / 92
PMSG	1800	20.5 / 92	18.2 / 96	15.5 / 92
SCIG	1200	18.1 / 92	16.6 / 92	16.1 / 96
SCIG	1500	26.4 / 92	23.5 / 92	23.8 / 92
SCIG	1800	51.0 / 100	47.2 / 100	44.0 / 96

Table B4 Sensitivity of MDE (%) (limit-of-detection convention, ‘-’ = largest extent not detectable)/DS (%) to the window length (3, 5, 8 cycles at $\alpha = 0.95$) and to the null quantile ($\alpha = 0.90$ and 0.99 at five cycles). ^s: the feature fails the reliability screen at that setting. IT: inter-turn; IW: inter-winding.

Class	Feature	Machine	3 c	5 c	8 c	$\alpha 0.90$	$\alpha 0.99$
IT	V_2/V_1	PMSG	3.8/73	3.8/76	3.8/73	3.8/80	3.8/69
IT	V_2/V_1	SCIG	7.4/69	7.4/73	7.4/75	7.4/76	7.4/65
IT	I_2/I_1	PMSG	7.4/57	7.4/56	7.4/58	7.4/65	7.4/52
IT	I_2/I_1	SCIG	7.4/70	7.4/75	7.4/75	7.4/79	7.4/68
IT	$I_d 2f_e$	PMSG	12.1/29	12.1/30	12.1/28	12.1/33	12.1/24
IT	$I_d 2f_e$	SCIG	7.4/78	7.4/79	7.4/83	7.4/84	7.4/72
IT	$PI_d 2f_e$	PMSG	12.1/30	12.1/31	12.1/28	12.1/33	12.1/24
IT	$PI_d 2f_e$	SCIG	7.4/78	7.4/78	7.4/83	7.4/84	7.4/72
IT	$V_q 2f_e$	PMSG	7.4/55	7.4/57	7.4/60	7.4/65	7.7/50
IT	$V_q 2f_e$	SCIG	7.4/61	7.4/60	7.4/62	7.4/64	7.4/59
IW	V_2/V_1	PMSG	9.7/92	2.3/94	2.3/94	2.3/94	9.7/91
IW	V_2/V_1	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	I_2/I_1	PMSG	9.7/83	9.7/81	9.7/82	3.2/87	9.7/79
IW	I_2/I_1	SCIG	2.3/98	2.3/100	2.3/100	2.3/100	2.3/97
IW	$I_d 2f_e$	PMSG	21.7/71	21.7/72	21.7/69	21.7/74	21.7/69
IW	$I_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$PI_d 2f_e$	PMSG	21.7/71	21.7/72	21.7/70	21.7/74	21.7/69
IW	$PI_d 2f_e$	SCIG	2.3/100	2.3/100	2.3/100	2.3/100	2.3/100
IW	$V_q 2f_e$	PMSG	9.7/89	9.7/90	9.7/91	9.7/91	9.7/86
IW	$V_q 2f_e$	SCIG	9.7/88	9.7/92	9.7/88	9.7/95	9.7/84

Table B5 Features failing the reliability screen (three or more false alarms on the nine healthy trials at $\alpha = 0.95$; count in parentheses) per window length and machine.

Window (cycles)	Machine	Excluded features (false alarms)
3	SCIG	D_2/D_1 (3)
5	PMSG	$I_d 1f_e$ (3), speed std (4)
5	SCIG	$I_a h7$ (3), D_2/D_1 (4)
8	PMSG	PI_d std (3), speed std (4)
8	SCIG	$I_q 2f_e$ (4), D_2/D_1 (5), V_{dc} std (4)

Table B6 Transfer of a GBDT detector restricted to each suite’s features (AUC), PMSG and SCIG, five-cycle windows: features z -scored on the target’s healthy trials (left block) and $|z|$ against each trial’s own reference (right block). P: PMSG, S: SCIG; same-machine columns use five-fold cross-validation grouped by tap pair.

Suite	z on target’s healthy trials				$ z $ vs own reference			
	P→P	S→S	P→S	S→P	P→P	S→S	P→S	S→P
drive-internal	0.87	0.87	0.73	0.74	0.89	0.95	0.92	0.86
3 CT	0.85	0.87	0.74	0.62	0.86	0.90	0.91	0.82
3 CT + 3 VT	0.84	0.89	0.86	0.70	0.89	0.90	0.92	0.89
3 CT + 3 VT + drive	0.88	0.90	0.81	0.75	0.92	0.95	0.92	0.86
+ torque transducer	0.88	0.90	0.82	0.75	0.92	0.95	0.92	0.86

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